

InnerVoice AI

EXPLORE THE EMOTIONAL PATTERNS BEHIND TEXT

Technical Report

SE - 801 - Software Project Lab III

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INSTITUTE OF INFORMATION TECHNOLOGY
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Letter of Transmittal

26th February, 2026

BSSE 4th Year Exam Committee
Institute of Information Technology
University of Dhaka

Subject: Technical Report Submission **InnerVoice AI** (Explore the emotional patterns behind text).

Sir,

I am submitting the technical report for our Software Project Lab 3, “**InnerVoice AI**: Explore the emotional patterns behind text.” This report comprehensively covers the project’s technical details, including design, implementation, and testing phases. While I’ve tried to do my best, I welcome your feedback for improvement.

Thank you for your consideration.

Sincerely,

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26.02.2026

Supervisor’s Signature

Acknowledgement

I am grateful to the Institute of Information Technology for giving me the opportunity to work on the project “**InnerVoice AI** (Explore the emotional patterns behind text).” I would like to convey my tremendous gratitude to my supervisor, Dr. Zerina Begum, Professor, Institute of Information Technology, University of Dhaka, for providing me with guidance and support throughout the project. Her expertise and insights were instrumental in shaping the direction of the project.

Lastly, I would like to thank my classmates. They have always been helpful and provided valuable insights from time to time.

Abstract

InnerVoice AI is an intelligent language analysis platform designed to help users better understand how their Facebook posts reflect emotional states and personality traits. In the modern digital landscape, social media has become a primary outlet for self-expression, yet users are often unaware of how their words may appear to others or what their language reveals about their inner psychological patterns.

Leveraging advances in Natural Language Processing (NLP) and AI, **InnerVoice AI** analyzes Bangla social media text to identify sentiment, emotional tone, and personality cues. The system maps linguistic patterns to the OCEAN personality model—Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism—providing users with meaningful, interpretable insights into their online expression.

Beyond analysis, **InnerVoice AI** promotes healthier communication by offering AI-powered rewriting suggestions when emotional imbalance, negativity, or harsh tones are detected. These suggestions help users reframe their thoughts in a more empathetic, balanced, and socially positive manner without altering the core message.

The platform also supports long-term emotional and personality tracking, allowing users to observe changes and trends over time, such as improvements in agreeableness or emotional stability. By addressing the research gap in Bangla social media text analysis, **InnerVoice AI** serves as a valuable tool for self-awareness, emotional well-being, and responsible digital communication for Bangla-speaking users.

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Chapter 1

Introduction

Social media, especially Facebook, has become a common place for people to share their thoughts and emotions. In Bangladesh, most of these expressions are written in Bangla and often reflect users' feelings, attitudes, and personal values. However, people are not always aware of how their posts may appear to others. What seems normal or emotional to the writer can sometimes be misunderstood as negative or aggressive.

Language often reveals emotions and personality traits without the writer realizing it. Although social media strongly affects mental well-being and public perception, tools for analyzing Bangla text are still limited. **InnerVoice AI** aims to fill this gap by analyzing Bangla Facebook posts to identify sentiment, emotional tone, and personality cues, helping users become more self-aware and communicate more positively.

1.1 Purpose

The main purpose of **InnerVoice AI** is to help users understand the emotions and personality traits reflected in their Bangla Facebook posts. The system analyzes text to identify sentiment, emotional balance, and personality patterns based on the OCEAN model. It also suggests improved rewrites when a post seems overly negative or emotionally unbalanced. Additionally, **InnerVoice AI** allows users to track changes in their emotional and personality patterns over time, supporting self-awareness and personal growth.

1.2 Motivation

InnerVoice AI is motivated by the strong influence of social media language on mental health and social relationships. Many users unknowingly express stress or negativity through their posts, which can cause misunderstandings. While such tools exist for English, Bangla-focused solutions are still limited. This project aims to fill that gap by providing a simple, culturally relevant tool that promotes positive communication and advances research in Bangla language analysis.

1.3 Scope

InnerVoice AI is designed to perform the following functions:

1. Analyze Bangla Facebook posts to determine sentiment (positive, negative, or neutral).
2. Identify underlying emotions such as joy, anger, sadness, fear, and surprise.
3. Map linguistic patterns to personality traits based on the OCEAN model.
4. Provide AI-assisted rewriting suggestions to promote empathetic and balanced expression.
5. Track emotional and personality trends over time and present progress in an understandable manner.

1.4 Timeline

Below is the proposed timeline for the **InnerVoice AI** project in a Gantt chart format:

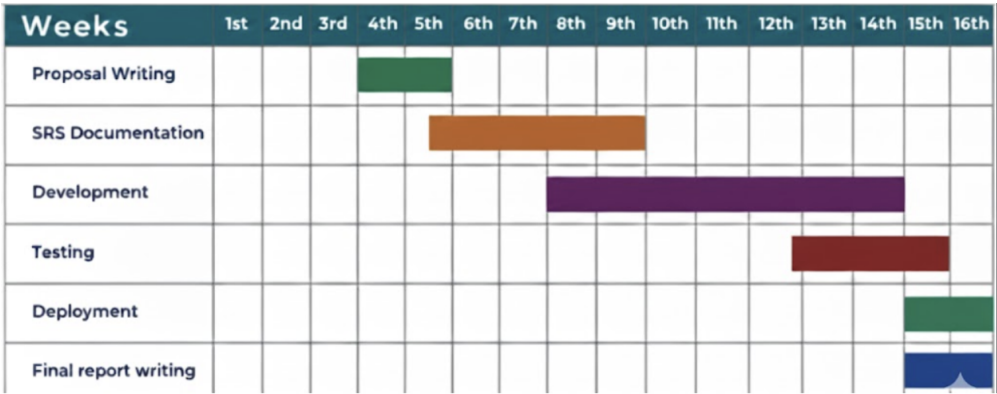


Figure 1.1: Project Timeline - Gantt Chart

1.5 Purpose of Document

The purposes of this document are:

- Identify and analyze the requirements
- Design the architecture
- Design the test plan

- Describe methodology
- Reduce the development effort
- Improve understanding

Chapter 2

Project Description

InnerVoice AI is designed to help people better understand how their Bangla Facebook posts reflect their emotions and personality. On social media, people often write impulsively, without thinking about how their words may sound to others. These posts can unintentionally appear negative, aggressive, or emotionally unstable. **InnerVoice AI** aims to bring awareness to this issue by analyzing written content and highlighting the emotional and psychological signals hidden within the text.

The system uses natural language processing techniques to examine Bangla posts and identify overall sentiment and specific emotions such as happiness, anger, sadness, fear, or surprise. Based on language patterns, **InnerVoice AI** also provides personality insights using the OCEAN model. When a post seems emotionally unbalanced or overly negative, the system suggests improved versions of the text that sound more calm, positive, and empathetic. Over time, users can also see how their emotional expression and personality indicators change, helping them develop healthier communication habits.

2.1 Requirements

2.1.1 Normal Requirements

Normal requirements describe the basic features that **InnerVoice AI** must have to work effectively.

Functional Requirements

- **Sentiment Detection:** Analyze Bangla posts to determine whether the tone is positive, negative, or neutral.
- **Emotion Identification:** Detect common emotions such as joy, anger, sadness, fear, and surprise.
- **Personality Insight:** Provide simple personality indicators based on the OCEAN model.

- **Text Rewriting Support:** Suggest alternative ways to rewrite posts when strong negativity or emotional imbalance is found.

Non-Functional Requirements

- **Ease of Use:** The system should be simple to use, allowing users to paste text and view results without difficulty.
- **Fast Response:** Analysis results should be generated within a short time.

2.1.2 Expected Requirements

Expected requirements are features users naturally assume the system will provide.

Functional Requirements

- **Emotional Consistency:** Ensure that emotional analysis and personality insights make sense together.
- **Clear Presentation:** Show results in an easy-to-understand format, such as summaries or simple charts.
- **Informal Language Handling:** Correctly analyze casual Bangla, slang, and everyday expressions used on Facebook.

Non-Functional Requirements

- **Reliability:** The system should work consistently without frequent errors.

2.1.3 Exciting Requirements

Exciting requirements are additional features that make the system more engaging and useful.

Functional Requirements

- **Progress Tracking :** For users who submit posts regularly, InnerVoice AI stores past results and displays changes over time, helping users notice improvements or recurring emotional patterns.

2.2 Usage Scenario for InnerVoice AI

InnerVoice AI is a web-based platform intended for everyday users who want to understand their online behavior better.

2.2.1 Access

Users visit the **InnerVoice AI** website and either use the system directly.

2.2.2 Text Submission and Analysis

Users paste a Bangla Facebook post into the input box. The system then analyzes the text to identify sentiment, emotions, and personality-related patterns.

2.2.3 Feedback and Suggestions

After processing, **InnerVoice AI** shows:

- The emotional tone of the post
- Detected emotions
- Personality indicators based on language use

If the post seems emotionally negative, the system offers rewritten versions that express the same idea in a calmer and more positive way.

Example Scenario:

A user writes a frustrated Facebook post after a stressful day. **InnerVoice AI** detects high negativity and emotional tension and suggests a softer rewrite. Over time, the user notices that their emotional balance has improved, encouraging more thoughtful online expression.

Chapter 3

Scenario-Based Modeling

Scenario-based modeling is an approach used in software development and design to create models that depict how a system behaves or reacts in various real-world situations or scenarios. These scenarios are typically based on user interactions, events, or conditions and help developers understand and visualize how the software will function in different contexts. By simulating these scenarios, developers can better identify potential issues, make informed design decisions, and ensure that the software meets user requirements across different usage contexts.

3.1 Use Case Diagram

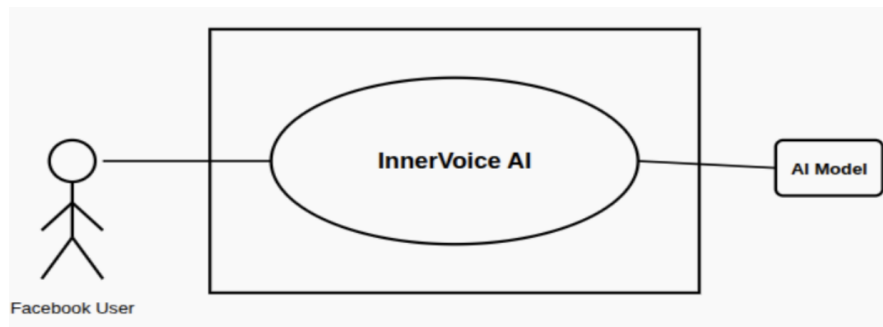
A use case diagram is a visual representation in software engineering that illustrates how a system interacts with its external entities or actors to achieve specific goals or tasks. It shows the relationships between use cases (representing system functionalities) and actors (representing external users or systems). Use case diagrams are typically created through requirements gathering, where system functionalities and user interactions are identified and then diagrammed using specialized software modeling tools.

3.1.1 Level 0 (InnerVoice AI)

Primary Actor: Facebook User

Secondary Actor: AI Model

Goal in Context: The diagram represents **InnerVoice AI**, a Bangla Facebook Post Analyzer that helps users understand and improve their social media expressions through sentiment analysis, emotion recognition, and personality insights.

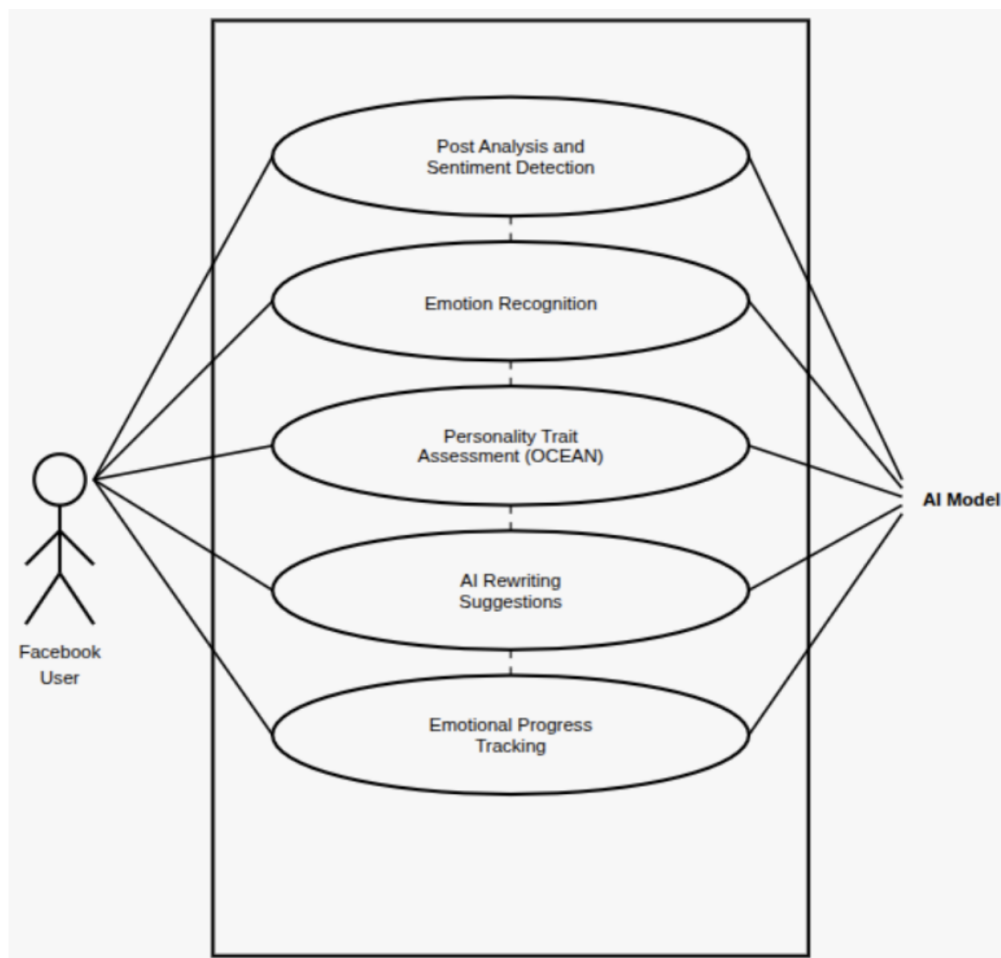
Figure 3.1: Use Case Level 0 (**InnerVoice AI**)

3.1.2 Modules of Level 1 (**InnerVoice AI**)

Primary Actor: Facebook User

Secondary Actor: AI Model

Goal in Context: The diagram illustrates the modular architecture of **InnerVoice AI**, showing how Facebook users interact with five primary functional modules, all supported by AI models that power the analytical and generative capabilities of the system.

Figure 3.2: Use Case Level 1 (**InnerVoice AI**)

Post Analysis and Sentiment Detection

This module is the main part of **InnerVoice AI** that analyzes user posts. Users can either paste their Bangla Facebook posts manually or upload text from their account. The system examines the text to understand the overall tone and determines whether it is positive, negative, or neutral. **InnerVoice AI** assigns a sentiment score to each post to show how strong the emotion is. It also highlights parts of the text that influence the result and explains why a certain sentiment was detected. The results are shown using simple visual tools such as sentiment indicators and highlighted words, making them easy for users to understand.

Emotion Recognition

The Emotion Recognition module goes deeper than basic sentiment analysis by identifying specific emotions expressed in a post. It detects emotions such as happiness, anger, sadness, fear, surprise, and disgust.

Each detected emotion is shown with an intensity level, helping users understand how strongly that emotion appears in their writing. The results are displayed using color-based visuals and percentage breakdowns, allowing users to see if their post contains a mix of emotions rather than just one.

Personality Trait Assessment (OCEAN Model)

This module analyzes writing patterns to provide insights into personality traits using the OCEAN model. By reviewing language style, word choices, and sentence structure, **InnerVoice AI** estimates traits such as openness, conscientiousness, extraversion, agreeableness, and neuroticism.

The system becomes more accurate as more posts are analyzed over time. Personality results are shown using simple charts and bars, allowing users to easily see which traits are more dominant and how they change with continued usage.

AI Rewriting Suggestions

When a post appears too negative, harsh, or emotionally unbalanced, this module offers rewriting suggestions. The AI generates alternative versions of the post that keep the original meaning but improve tone and clarity.

The system highlights words or phrases that may cause negative impressions and explains why suggested changes might sound more positive or empathetic. Users can compare the original and rewritten versions side by side and choose the one they prefer or modify it further.

Emotional Progress Tracking

This module helps users track their emotional and personality changes over time.

InnerVoice AI stores analysis results from previous posts and presents them in easy-to-read graphs and summaries.

Users can see trends such as improvements in emotional balance or reductions in negative tone over weeks or months. The system also provides simple progress messages, such as increases in agreeableness or decreases in negative sentiment. This long-term tracking encourages self-awareness and supports continuous personal growth.

Chapter 4

Data-Based Modeling

No database or data storage system is used for this tool. So this section does not contain any relevant analysis.

Chapter 5

Class-Based Modeling

Class-based modeling is a visual representation in software engineering that illustrates the structure of a software system using classes, their attributes, and relationships. It is commonly used in object-oriented design and modeling.

5.1 Analysis Classes

After analyzing the **InnerVoice AI** scenario, relevant nouns were identified from the problem description. These nouns were then filtered using general classification, including external entities, things, events, and system components. Further refinement was done using selection criteria, such as retained information, required services, multiple attributes, common operations, and essential system responsibilities.

The purpose of this analysis was to identify core classes that represent the internal structure of **InnerVoice AI**, which analyzes Bangla Facebook posts to extract sentiment, emotional tone, and personality traits and provide rewriting suggestions—without relying on any database system.

5.1.1 Identified Key Nouns from the Scenario

From the **InnerVoice AI** description, the following key nouns were identified as potential classes:

1. User
2. Facebook Post
3. Sentiment
4. Emotion
5. Personality
6. Rewrite Suggestion
7. Analysis Result

8. Text Analysis
9. Emotional Feedback
10. Personality Insight

5.1.2 Class Filtering and Merging

During analysis, it was observed that several nouns represent closely related responsibilities. For example:

- Sentiment, Emotion, and Personality are all analytical processes applied to text.
- Text Analysis acts as a coordination process rather than a standalone entity.
- Emotional Feedback and Personality Insights are forms of analysis output.

Based on overlapping responsibilities and to maintain a simple and modular design, similar concepts were grouped and represented through specialized analyzer classes.

5.1.3 Derived Classes and Final Grouping

After applying General Classification and Selection Criteria, the following seven analysis classes were finalized for **InnerVoice AI**:

1. User
2. Post
3. SentimentAnalyzer
4. EmotionAnalyzer
5. PersonalityAnalyzer
6. RewriteSuggestion
7. AnalysisResult

5.2 Class Cards

Table 5.1: CRC Card (User)

User
Attributes - userId - inputText
Methods + submitPost()
Responsibilities • Submit Bangla Facebook posts for analysis • View analysis results
Collaborators • Post

Table 5.2: CRC Card (Post)

Post
Attributes - postId - postContent - language
Methods - preprocessText() + sendForAnalysis()
Responsibilities • Store the user's submitted text • Prepare text for analysis
Collaborators • User • AnalysisController

Table 5.3: CRC Card (SentimentAnalyzer)

SentimentAnalyzer
Attributes - sentimentLabel - sentimentScore
Methods + analyzeSentiment() + calculateScore()
Responsibilities • Determine the overall tone of the text • Classify sentiment as positive, negative, or neutral
Collaborators • AnalysisController

Table 5.4: CRC Card (EmotionAnalyzer)

EmotionAnalyzer
Attributes - detectedEmotions - emotionIntensity
Methods + detectEmotion() + generateEmotionReport()
Responsibilities • Identify emotions expressed in the text • Measure emotional intensity
Collaborators • AnalysisController

Table 5.5: CRC Card (PersonalityAnalyzer)

PersonalityAnalyzer
Attributes - openness - conscientiousness - extraversion - agreeableness - neuroticism
Methods + analyzePersonality() + generatePersonalityProfile()
Responsibilities • Analyze personality traits using the OCEAN model • Generate a personality profile from text
Collaborators • AnalysisController

Table 5.6: CRC Card (RewriteSuggestion)

RewriteSuggestion
Attributes - originalText - suggestedText
Methods + generateRewrite() + compareTexts()
Responsibilities • Suggest an improved and emotionally balanced text • Reduce negative or harsh expressions
Collaborators • AnalysisController • Post

Table 5.7: CRC Card (AnalysisController)

AnalysisController
Attributes + post - sentimentResult - emotionResult - personalityResult
Methods + startAnalysis() + collectResults() + displayOutput()
Responsibilities • Coordinate all analysis modules • Present final results to the user
Collaborators • Post • SentimentAnalyzer • EmotionAnalyzer • PersonalityAnalyzer • RewriteSuggestion

5.3 CRC Diagram

The CRC diagram of the classes is given below:

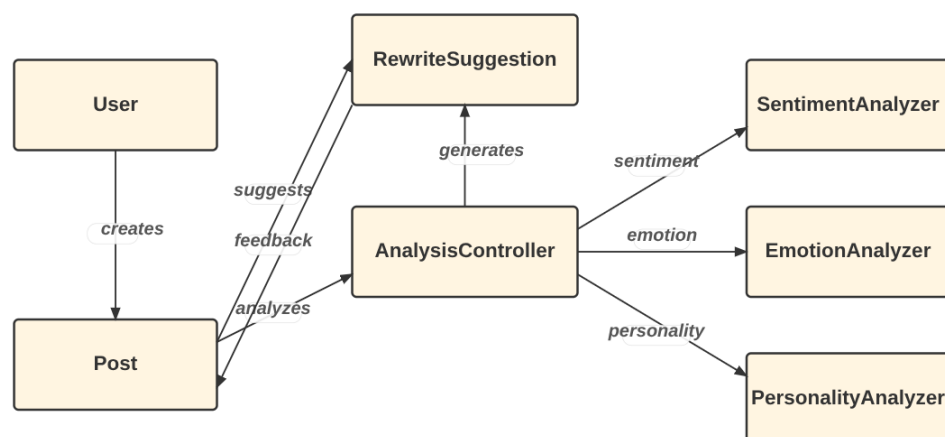


Figure 5.1: CRC Diagram of InnerVoice AI

Chapter 6

Architectural Design

Architectural design represents the high-level structure of a software system and explains how its main components are organized and interact with one another. It focuses on defining system boundaries, external entities, and communication interfaces rather than internal implementation details.

For **InnerVoice AI**, the architectural design highlights how the system processes Bangla Facebook posts to generate sentiment, emotion, personality insights, and rewriting suggestions.

6.1 Architectural Context Diagram

At the architectural design level, an Architectural Context Diagram (ACD) is used to illustrate how **InnerVoice AI** interacts with external entities beyond its system boundary. The diagram identifies external systems and actors that exchange information with the target system through defined interfaces.

External entities interacting with **InnerVoice AI** are classified as follows:

- **Superordinate systems:** Systems that use InnerVoice AI as part of a higher-level workflow.
- **Subordinate systems:** Systems or services that InnerVoice AI depends on to perform its core functionality.
- **Peer-level systems:** Systems that interact with InnerVoice AI on an equal level by exchanging data.
- **Actors:** Individuals or devices that directly interact with the system by providing inputs or receiving outputs.

6.1.1 Representation of InnerVoice AI in the Architectural Context Diagram

The architectural context of **InnerVoice AI** is described below:

- **Superordinate systems:** **InnerVoice AI** does not operate as a component of any higher-level system. Therefore, it has no superordinate systems.
- **Subordinate systems:** **InnerVoice AI** relies on Natural Language Processing (NLP) models and AI language models to analyze Bangla text. These models are used for sentiment detection, emotion recognition, personality assessment, and text rewriting. They function as subordinate systems by providing essential processing capabilities.
- **Peer-level systems:** **InnerVoice AI** does not directly interact with any peer-level systems. All analysis and feedback are handled within the system boundary.
- **Actors:** The primary actors of **InnerVoice AI** are social media users who submit Bangla Facebook posts for analysis. These users interact with the system to receive emotional insights, personality indicators, and rewriting suggestions.

6.1.2 Interface Description

Actors communicate with **InnerVoice AI** through a user interface that allows text submission and displays analysis results. Subordinate AI and NLP services interact with **InnerVoice AI** through internal processing interfaces that enable text analysis and feedback generation.

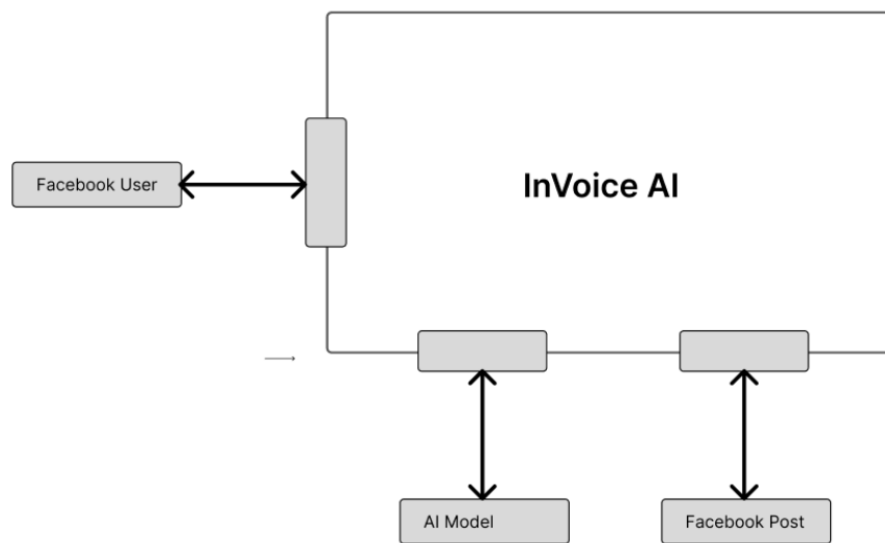


Figure 6.1: Architectural Context Diagram

6.2 Archetypes

The archetypes for the tool are defined below:

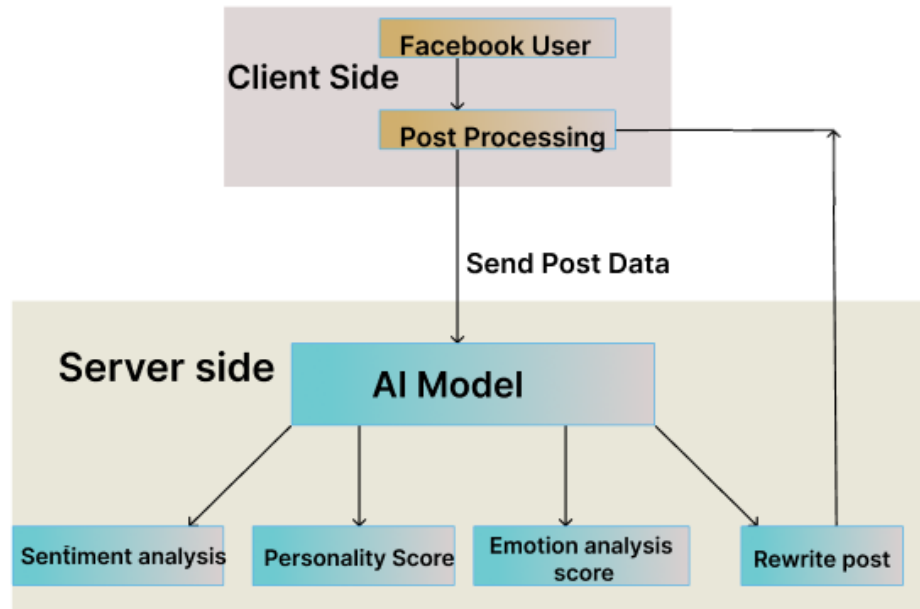


Figure 6.2: High-level overview of archetypes

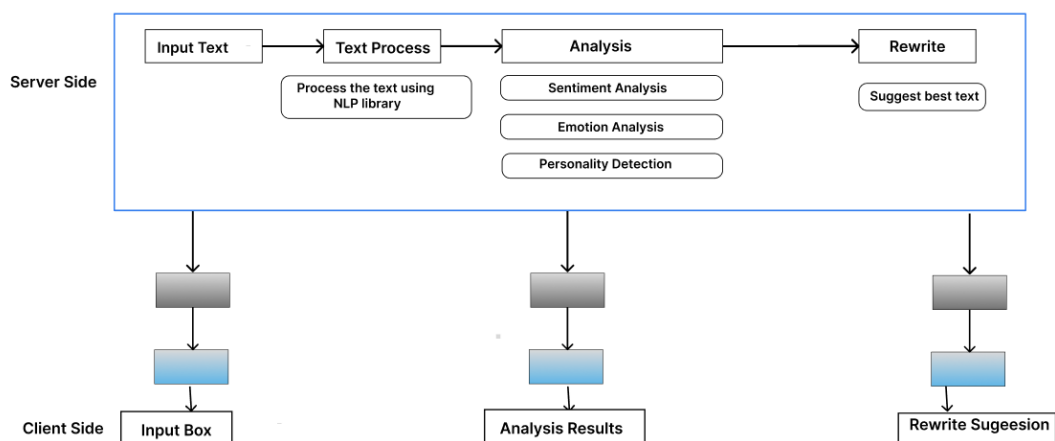


Figure 6.3: Archetype

6.2.1 Text Process (Server-side)

SProcess User Text Using NLP:

- Receives raw text input from client-side Input Box

- Applies NLP libraries for text preparation
- Handles Bangla and English language processing
- Converts unstructured text into machine-readable format for analysis

6.2.2 Text Analysis (Server-side)

Perform Text Intelligence Analysis:

- Executes Sentiment Analysis to determine emotional polarity
- Performs Emotion Analysis to detect feelings such as anger, happiness, sadness, etc.
- Applies Personality Detection models to infer user behavior pattern
- Uses trained AI/ML models optimized for social media text
- Generates structured analytical results for frontend consumption

6.2.3 Rewrite (Server-side)

Generate Optimized Text Suggestions:

- Uses AI-based text generation models to rewrite user content
- Considers sentiment, emotion, and personality analysis results
- Adjusts tone, clarity, and positivity of the original text
- Produces context-aware and user-friendly rewritten text
- Sends best rewrite suggestions back to client-side interface

6.2.4 Input Box (Client-side)

Capture User Text Input:

- Allows users to input Facebook posts
- Validates input length and format
- Sends text to server for processing

6.2.5 Analysis Results (Client-side)

Display Analysis Output:

- Shows sentiment, emotion, and personality results visually
- Helps users understand post impact

6.2.6 Rewrite Suggestion (Client-side)

Present AI-Generated Text:

- Displays rewritten post suggestions
- Enables copy and reuse functionality

6.3 Top Level Components

The overall architectural structure with top-level components is illustrated below:

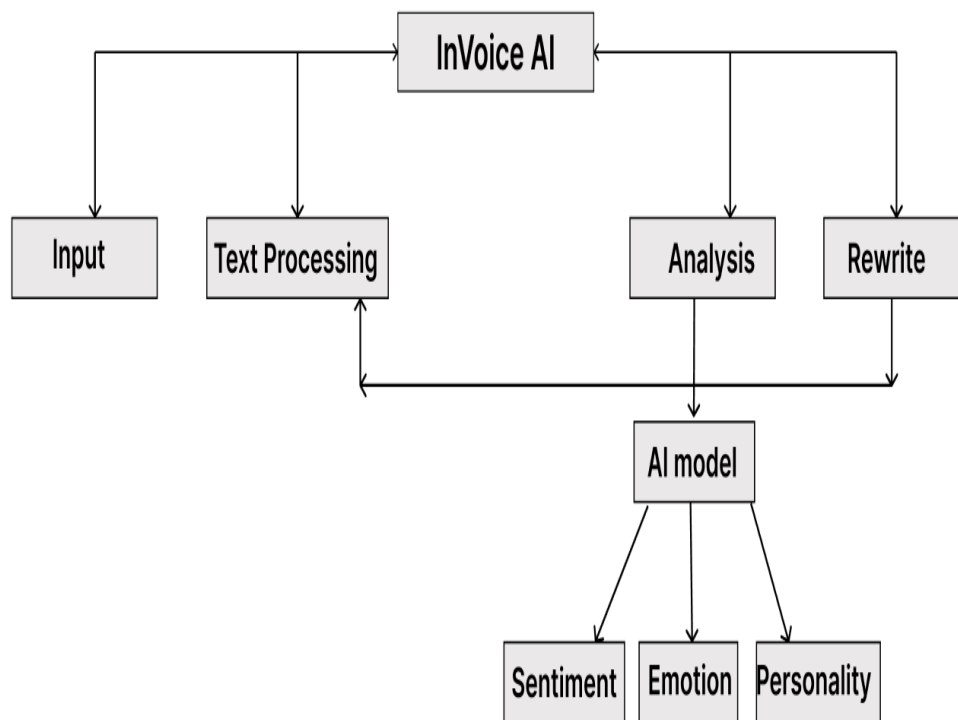


Figure 6.4: Top Level Components

6.4 Mapping Requirements to Software Architecture

The mapping of the derived components with the requirements can be shown below:

Table 6.1: Requirements to Architecture Mapping

Requirement	Architectural Components
Accept Bangla Facebook post input	User Interface, Post Component
Validate text input (non-empty, Bangla text)	User Interface, Post Component
Perform sentiment analysis	SentimentAnalyzer, NLP Service
Generate sentiment score and label	SentimentAnalyzer, AnalysisResult
Detect emotional states from text	EmotionAnalyzer, NLP Service
Identify emotion intensity	EmotionAnalyzer, AnalysisResult
Analyze personality traits using OCEAN model	PersonalityAnalyzer, NLP Service
Generate personality insight visualization	PersonalityAnalyzer, AnalysisResult
Provide AI-based rewriting suggestions	RewriteSuggestion, NLP Service
Preserve original meaning in rewrites	RewriteSuggestion
Display combined analysis results	AnalysisResult, User Interface
Support real-time analysis	AnalysisController, Analyzer Components
Allow reanalysis of modified text	Post Component, AnalysisController
Handle invalid input or processing errors	User Interface, AnalysisController
Ensure system stability during use	AnalysisController

Chapter 7

Applied Methodology

This chapter elucidates the internal mechanisms of **InnerVoice AI** (an AI-based Facebook post analyzer and rewrite suggester), detailing its workflow, input handling, processing modules, and output generation. By dissecting these components, we provide a comprehensive understanding of how **InnerVoice AI** operates to deliver actionable post analysis and rewriting capabilities.

7.1 System Workflow Overview

InnerVoice AI follows a structured pipeline that processes user-submitted Bangla Facebook posts through multiple analytical stages. The workflow can be summarized as follows:

1. User submits a Bangla Facebook post through the web interface
2. Input validation and preprocessing
3. Text analysis (sentiment, emotion, personality)
4. AI-based rewriting suggestion generation
5. Result compilation and visualization
6. Output presentation to the user

7.1.1 Workflow Diagram

The complete system workflow is illustrated in the following diagram:

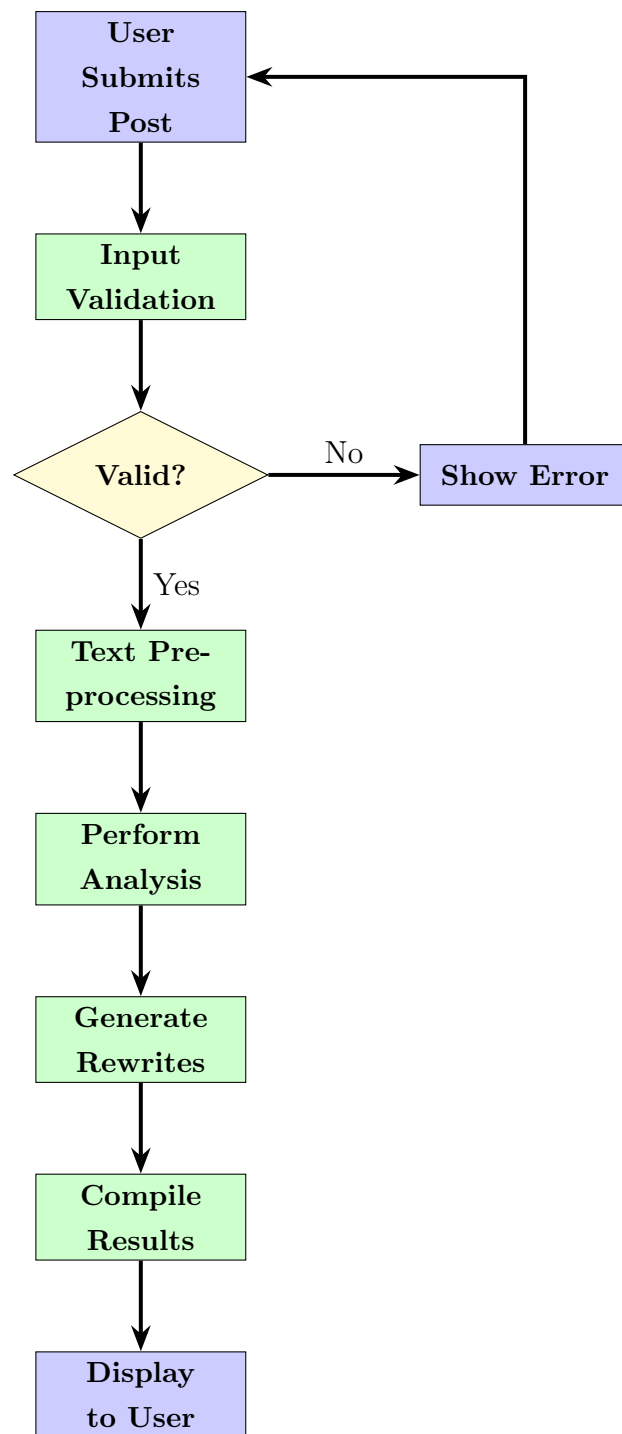


Figure 7.1: Complete System Workflow Diagram

7.2 Input Handling and Validation

7.2.1 Text Input Reception

Users interact with **InnerVoice AI** through a user-friendly web interface. The system accepts text input in the following manner:

- Users paste or type their Bangla Facebook posts into a designated text input area
- The system supports posts of varying lengths (minimum 10 characters, maximum 5000 characters)
- Input is received in real-time and immediately forwarded for validation

7.2.2 Input Validation

Before processing, the system performs comprehensive validation:

- **Non-Empty Check:** Ensures the input contains meaningful text
- **Language Detection:** Verifies that the input primarily contains Bangla or English text
- **Length Validation:** Confirms the text meets minimum and maximum length requirements
- **Character Encoding:** Ensures proper UTF-8 encoding for Bangla script support
- **Error Messaging:** Provides clear feedback if validation fails

7.3 Text Preprocessing

Before analysis, the input text undergoes preprocessing to prepare it for machine learning models:

7.3.1 Preprocessing Pipeline

The following diagram shows the sequential preprocessing stages:

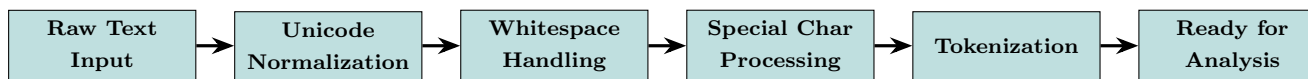


Figure 7.2: Text Preprocessing Pipeline

7.3.2 Text Normalization

- **Unicode Normalization:** Standardizes Bangla character representations

- **Whitespace Handling:** Removes excess spaces and normalizes line breaks
- **Special Character Handling:** Processes emojis and special symbols appropriately
- **Case Handling:** Preserves case for context-sensitive analysis

7.3.3 Tokenization

- **Sentence Splitting:** Divides text into sentences for granular analysis
- **Word Tokenization:** Breaks sentences into individual tokens
- **Bangla-Specific Processing:** Handles Bangla morphology and compound words

7.4 Core Analysis Modules

7.4.1 Analysis Modules Architecture

The following diagram illustrates how the three core analysis modules process the input text in parallel:

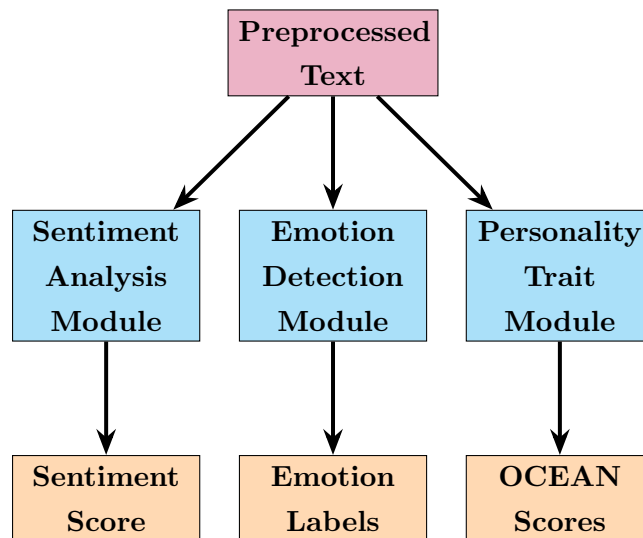


Figure 7.3: Core Analysis Modules Architecture

7.4.2 Sentiment Analysis Module

The Sentiment Analysis module determines the overall emotional polarity of the post.

Methodology:

- Utilizes pre-trained transformer-based models optimized for Bangla text
- Analyzes linguistic patterns, word choices, and contextual cues
- Generates a sentiment score ranging from -1.0 (negative) to +1.0 (positive)
- Classifies posts into three categories: Positive, Negative, or Neutral
- Provides confidence scores for each classification

7.4.3 Emotion Detection Module

The Emotion Detection module identifies specific emotions expressed in the text.

Methodology:

- Applies multi-label emotion classification models
- Detects emotions including: Joy, Anger, Sadness, Fear, Surprise, and Disgust
- Assigns intensity levels to each detected emotion
- Handles mixed emotions and emotional ambiguity through probabilistic scoring
- Generates emotion intensity percentages for visualization

7.4.4 Personality Trait Assessment Module

The Personality Trait Assessment module analyzes linguistic patterns to infer personality traits using the OCEAN model.

Methodology:

- **Openness:** Evaluates language creativity, use of novel words, and abstract thinking
- **Conscientiousness:** Assesses organization indicators, attention to detail, and planning language
- **Extraversion:** Analyzes social orientation, activity level descriptions, and engagement patterns
- **Agreeableness:** Evaluates cooperative language, empathy indicators, and social harmony cues

- **Neuroticism:** Measures emotional volatility, stress indicators, and negativity patterns
- Generates OCEAN profile scores normalized between 0 and 100
- Creates personality insights based on dominant traits

7.5 Rewriting Suggestion Generation

7.5.1 Rewrite Process Flow

The following diagram illustrates how the system detects negative content and generates alternative suggestions:

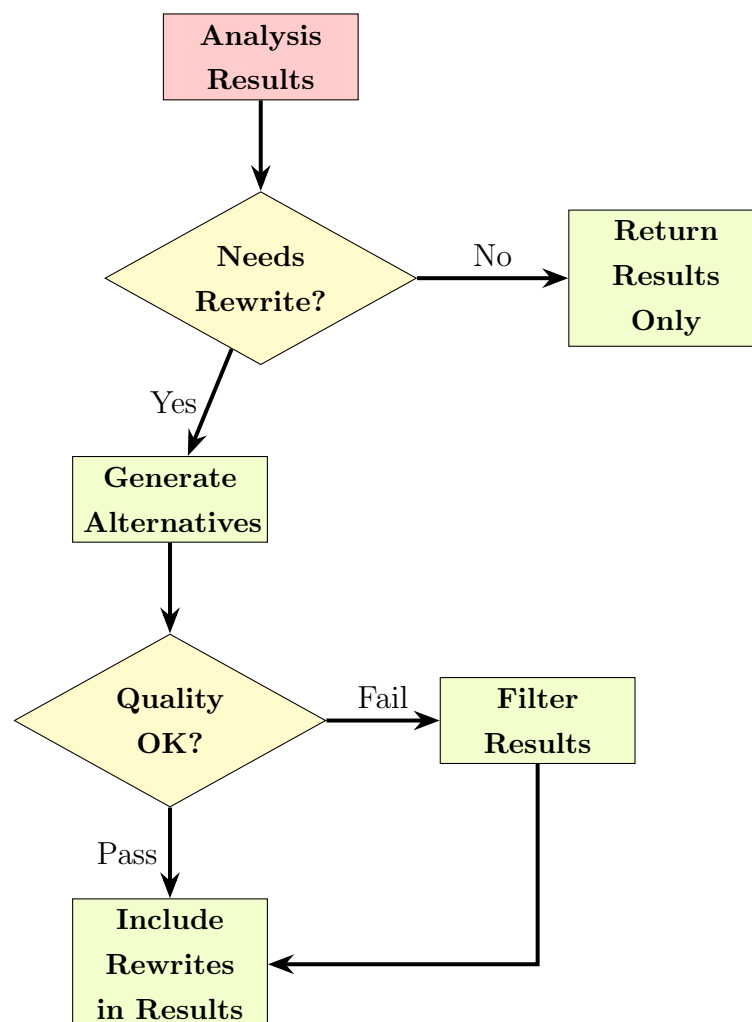


Figure 7.4: Rewriting Suggestion Generation Flow

7.5.2 Negative Content Detection

The system identifies posts that may benefit from rewriting:

- Detects overly negative sentiment scores (below -0.3)
- Identifies emotionally imbalanced content (high anger or sadness)
- Flags harsh or aggressive language patterns
- Recognizes posts with potential communication issues

7.5.3 AI-Based Text Rewriting

When negative or imbalanced content is detected, the system generates rewriting suggestions:

- Utilizes advanced language models (such as transformative models) for text generation
- Maintains semantic meaning and original intent
- Improves emotional tone and clarity
- Generates multiple alternative suggestions (typically 2-3 options)
- Highlights specific words or phrases that were modified
- Explains the rationale behind suggested changes

7.5.4 Rewrite Quality Assurance

- Validates that rewrites preserve core message
- Ensures readability and natural language flow
- Checks for length constraints and formatting consistency
- Filters out inappropriate or nonsensical suggestions

7.6 Result Compilation and Integration

7.6.1 Result Aggregation Architecture

The following diagram shows how all analysis results are combined:

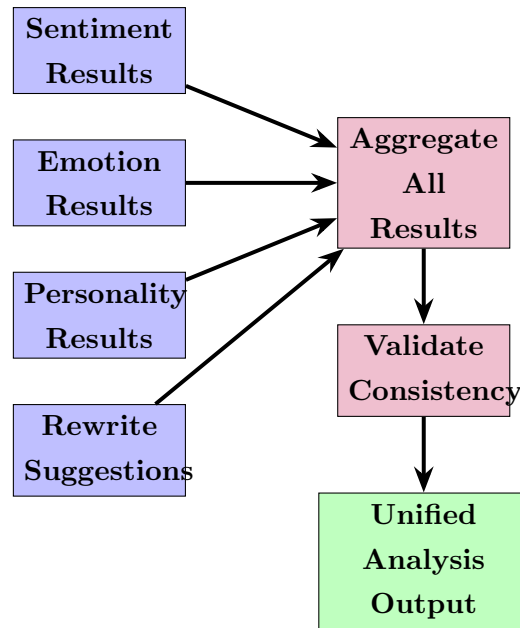


Figure 7.5: Result Compilation and Integration

7.6.2 Result Aggregation

All analysis results are compiled into a unified structure:

- Combines sentiment analysis results
- Integrates emotion detection findings
- Incorporates personality trait assessments
- Includes rewriting suggestions (if applicable)
- Adds metadata such as analysis timestamps and confidence scores

7.6.3 Result Validation

- Ensures consistency between different analytical modules
- Validates emotional and sentiment alignment
- Checks for logical consistency in personality assessments
- Handles conflicts or inconsistencies through scoring mechanisms

7.7 Output Generation and Visualization

7.7.1 User Interface Output Structure

The following diagram illustrates how analysis results are presented to users:

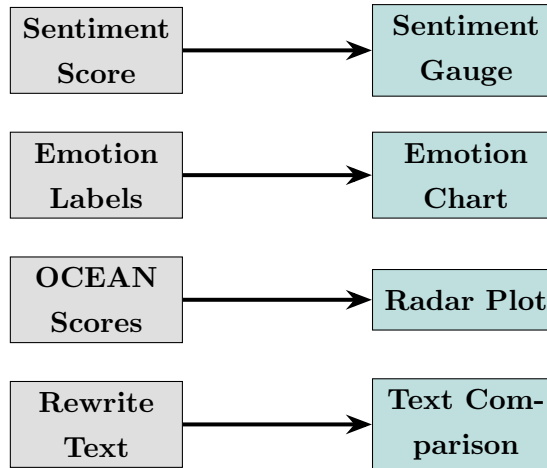


Figure 7.6: Output Generation and Visualization

7.7.2 User Interface Presentation

The compiled results are presented to users in an intuitive, visually organized format:

- **Sentiment Display:** Visual gauge showing sentiment polarity and score
- **Emotion Breakdown:** Bar chart or circular visualization of detected emotions
- **Personality Profile:** OCEAN model representation using radar charts or bar graphs
- **Rewriting Suggestions:** Side-by-side comparison of original and suggested text
- **Detailed Insights:** Text-based explanations of key findings

7.7.3 Interactive Features

- Users can modify original text and request reanalysis
- Copy-to-clipboard functionality for rewritten suggestions
- Expandable sections for detailed explanations
- Color-coded indicators for quick visual understanding

7.8 Technology Stack

InnerVoice AI utilizes the following technologies and frameworks:

7.8.1 Backend Technologies

- **Framework:** Django (Python web framework)
- **NLP Libraries:** NLTK, spaCy, and Hugging Face Transformers
- **Machine Learning:** scikit-learn, TensorFlow/Keras
- **APIs:** Google Gemini API for advanced text generation
- **Database:** SQLite for user data management (optional)

7.8.2 Frontend Technologies

- **Framework:** React.js / Vue.js
- **Styling:** CSS3, Tailwind CSS, or Material-UI
- **Visualization:** Chart.js or D3.js for graphical representations
- **HTTP Client:** Axios or Fetch API for backend communication

7.9 Processing Pipeline Architecture

The complete processing pipeline follows this sequence:

User Input ➤ Validation ➤ Preprocessing ➤ Analysis ➤ Rewriting ➤ Compilation ➤ Visualization

Figure 7.7: InnerVoice AI Processing Pipeline

7.10 Error Handling and Robustness

7.10.1 Error Management

- **Input Errors:** Clear messages for invalid or unsupported input
- **Processing Errors:** Graceful fallback mechanisms if analysis modules fail
- **Model Errors:** Timeout handling and fallback to simpler analysis if complex models fail
- **Network Errors:** Retry mechanisms for external API calls

7.10.2 Fallback Strategies

- If advanced emotion detection fails, system falls back to basic sentiment analysis
- If rewriting generation times out, system notifies user instead of hanging
- If personality assessment is inconclusive, system indicates uncertainty in results

7.11 Performance Optimization

InnerVoice AI implements several optimization strategies:

- **Model Caching:** Pre-loads trained models into memory for faster inference
- **Batch Processing:** Groups multiple requests when possible
- **Asynchronous Processing:** Handles long-running tasks without blocking the UI
- **Result Caching:** Stores recent analysis results for quick retrieval

Chapter 8

Test Planning

This chapter presents the test plan for **InnerVoice AI**, structured in accordance with the IEEE 829 Standard for Software Test Documentation and aligned with ISO/IEC 29119 testing practices. The plan defines the scope, objectives, strategy, environment, entry/exit criteria, and detailed test cases that govern the verification and validation of all system components.

8.1 Test Plan Overview

8.1.1 Purpose

The purpose of this test plan is to:

1. Define the scope and objectives of testing activities for **InnerVoice AI**.
2. Establish a structured testing strategy across multiple test levels.
3. Specify entry and exit criteria for each testing phase.
4. Document detailed test cases with preconditions, input data, and expected outcomes.
5. Provide a requirements-to-test traceability matrix to ensure complete coverage.

8.1.2 Scope

In Scope

- Language detection for Bangla (Unicode), Banglish (Roman-script), English, and mixed-language inputs
- Sentiment analysis (positive, negative, neutral classification with probability scores)

- Emotion detection (joy, sadness, anger, fear, surprise, neutral)
- Personality trait assessment using the OCEAN model
- AI-powered text rewriting suggestions
- RESTful API endpoints (authentication, analysis, rewrite, history, progress)
- Data model integrity (defaults, constraints, cascade behavior, ordering)
- End-to-end user workflows (registration, analysis, rewrite, data export, account deletion)
- Security controls (authentication enforcement, data isolation between users)

Out of Scope

- Frontend (Chrome Extension) user interface testing
- Load testing and performance benchmarking under concurrent users
- Penetration testing and advanced security auditing
- Third-party API reliability testing (Google Gemini uptime)

8.1.3 References

- IEEE 829-2008: Standard for Software and System Test Documentation
- ISO/IEC 29119-3: Software Testing — Test Documentation
- Django Testing Documentation: <https://docs.djangoproject.com/en/5.0/topics/testing/>
- Django REST Framework Testing: <https://www.django-rest-framework.org/api-guide/testing/>

8.2 Test Strategy

The testing strategy for **InnerVoice AI** adopts a **V-Model** approach, where each development phase has a corresponding testing phase. Testing is organized into four distinct levels, each targeting a specific layer of the system architecture.

8.2.1 Test Levels

The following diagram illustrates the test levels and their relationship to system components:

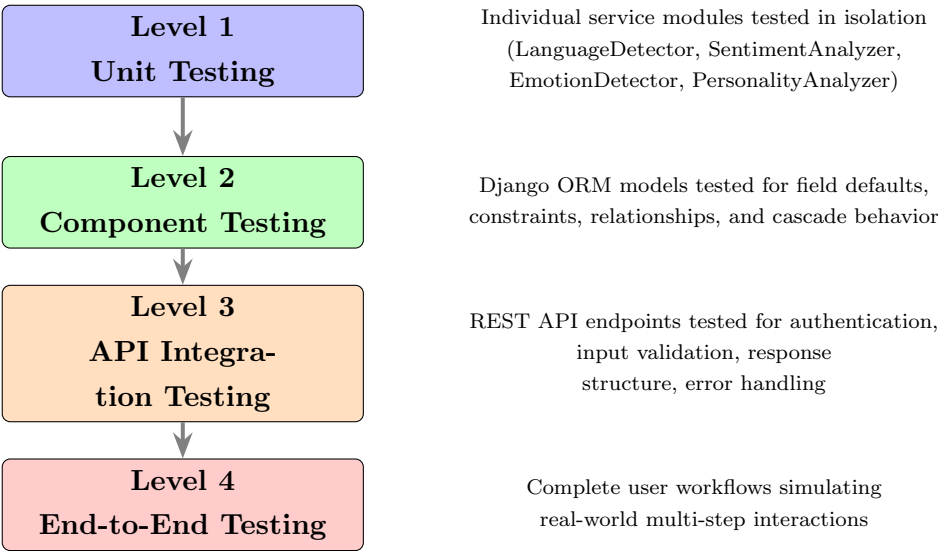


Figure 8.1: Test Levels Hierarchy

Table 8.1: Test Levels and Corresponding Techniques

Level	Test Type	Scope	Technique
1	Unit Testing	Individual NLP/AI service modules	White-box
2	Component Testing	Django ORM models and serializers	White-box
3	API Integration	REST endpoint request/response contracts	Black-box
4	End-to-End	Multi-step user workflows across modules	Black-box

8.2.2 Test Design Techniques

The following test design techniques are employed:

- **Equivalence Partitioning:** Input domains are divided into equivalence classes (e.g., valid Bangla text, valid English text, empty input, whitespace-only input) and one representative test case is selected from each partition.

- **Boundary Value Analysis:** Boundary conditions are tested (e.g., minimum text length of 3 characters, maximum text length, sentiment score boundaries at -1.0 , 0.0 , and $+1.0$).
- **Error Guessing:** Tests are designed for common failure modes such as `None` input, model loading failures, invalid UUIDs, and expired authentication tokens.
- **State Transition Testing:** Authentication state transitions are tested (unauthenticated $\rightarrow$ registered $\rightarrow$ logged in $\rightarrow$ logged out $\rightarrow$ token invalidated).
- **Decision Table Testing:** Rewrite eligibility logic is tested across combinations of sentiment score, emotion intensity, and language type.

8.3 Test Environment

Table 8.2: Test Environment Specification

Component	Specification
Operating System	Ubuntu Linux 22.04 LTS
Runtime	Python 3.10+
Framework	Django 4.x with Django REST Framework 3.x
Test Runner	Django <code>manage.py test</code> (unittest-based)
Test Database	SQLite (in-memory, auto-created per test class)
Mocking Library	<code>unittest.mock</code> (stdlib) with <code>@patch</code> decorators
HTTP Client	<code>rest_framework.test.APIClient</code>
Fixture Management	<code>tests/conftest.py</code> (shared factories and mock data)
CI/CD	Tests executable via <code>python manage.py test tests/</code>

8.4 Entry and Exit Criteria

8.4.1 Entry Criteria

Testing shall commence only when the following conditions are satisfied:

1. All source code for the module under test has been committed to the version control repository.
2. The development environment is correctly configured with all dependencies installed (`requirements.txt`).

3. The test database schema is auto-generated via Django migrations without errors.
4. Shared test fixtures in `conftest.py` are functional and return valid mock objects.
5. External service dependencies (Gemini API, ML models) have corresponding mock implementations.

8.4.2 Exit Criteria

A testing phase is considered complete when:

1. All planned test cases have been executed.
2. All test cases with **High** and **Medium** priority have passed.
3. No unresolved **Critical** or **High** severity defects remain open.
4. Test coverage meets the target threshold (all public API endpoints and service methods covered).
5. Test execution results have been documented and reviewed.

8.5 Test Cases

This section presents detailed test cases organized by test level. Each test case follows a structured format with test scenario, input/precondition, steps, expected outcomes, and result.

8.5.1 Level 1: Unit Test Cases — Service Modules

TC001 Language Detection — Bangla Unicode

Test Scenario	Verify that the <code>LanguageDetector</code> correctly identifies Bangla Unicode script input.
Input / Precondition	<code>LanguageDetector</code> class is available and instantiated.
Steps	1. Instantiate the <code>LanguageDetector</code>. 2. Pass the Bangla text ‘<code>বাংলা</code>’ to the detection method. 3. Check the returned <code>language_code</code>.
Expected Outcomes	– <code>language_code</code> is ‘bn’. – <code>is_supported</code> is True. – <code>script_type</code> is ‘bangla’.
Result	Passed — Bangla Unicode text correctly detected as ‘bn’.

TC002 Language Detection — Banglish (Roman-script Bangla)

Test Scenario	Verify that the <code>LanguageDetector</code> correctly identifies Banglish input written in Roman script.
Input / Precondition	<code>LanguageDetector</code> class is available and instantiated.
Steps	1. Instantiate the <code>LanguageDetector</code>. 2. Pass the Banglish text ‘<code>ami khub khushi ajke</code>’ to the detection method. 3. Check the returned <code>language_code</code>.
Expected Outcomes	– <code>language_code</code> is ‘bn’. – <code>script_type</code> is ‘banglish’.
Result	Passed — Banglish text correctly detected as Bangla.

TC003 Language Detection — Empty String Handling

Test Scenario	Verify that the <code>LanguageDetector</code> handles empty string input gracefully without crashing.
Input / Precondition	<code>LanguageDetector</code> class is available and instantiated.
Steps	1. Instantiate the <code>LanguageDetector</code>. 2. Pass an empty string <code>''</code> to the detection method. 3. Check the returned language code and supported flag.
Expected Outcomes	– Falls back to <code>language_code = 'en'</code>. – No exception is raised.
Result	Passed — Empty input handled gracefully with fallback to English.

TC004 Sentiment Analysis — Model Loading

Test Scenario	Verify that the <code>SentimentAnalyzer</code> model loads successfully at startup and produces a valid analyzer instance.
Input / Precondition	Python environment is configured with all dependencies installed.
Steps	1. Instantiate the <code>SentimentAnalyzer</code> class. 2. Check that the internal <code>self.analyzer</code> attribute is not <code>None</code>.
Expected Outcomes	– <code>self.analyzer</code> is not <code>None</code>. – No exception is raised during instantiation.
Result	Passed — Sentiment model loaded successfully.

TC005 Sentiment Analysis — Positive Sentiment Classification

Test Scenario	Verify that the <code>SentimentAnalyzer</code> correctly classifies strongly positive text as <code>positive</code> .
Input / Precondition	Sentiment model is loaded and ready.
Steps	1. Pass the text ‘I love this amazing wonderful day!’ to the analyzer. 2. Retrieve the sentiment label and score. 3. Verify the probability distribution sums to approximately 1.0.
Expected Outcomes	– label is ‘positive’. – score is a valid float. – Sum of <code>positive</code>, <code>neutral</code>, <code>negative</code> scores $\approx$ 1.0.
Result	Passed — Positive text classified correctly with valid scores.

TC006 Sentiment Analysis — Negative Sentiment Classification

Test Scenario	Verify that the <code>SentimentAnalyzer</code> correctly classifies strongly negative text as <code>negative</code> .
Input / Precondition	Sentiment model is loaded and ready.
Steps	1. Pass the text ‘I hate this terrible awful day’ to the analyzer. 2. Retrieve the sentiment label.
Expected Outcomes	– label is ‘negative’. – <code>negative</code> score is the highest among all categories.
Result	Passed — Negative text classified correctly.

TC007 Sentiment Analysis — Graceful Fallback on Model Failure

Test Scenario	Verify that the <code>SentimentAnalyzer</code> returns safe default values when the underlying model is unavailable.
Input / Precondition	Model is set to <code>None</code> to simulate loading failure.
Steps	1. Set the analyzer's model to <code>None</code>. 2. Pass the text '<code>Test text</code>' to the analyzer. 3. Check the returned result.
Expected Outcomes	– <code>label</code> is '<code>neutral</code>', <code>score</code> is 0.0. – Scores default to approximately equal distribution (≈ 0.33 each). – No exception is raised.
Result	Passed — Fallback values returned safely on model failure.

TC008 Emotion Detection — All Emotion Categories Present

Test Scenario	Verify that the <code>EmotionDetector</code> output contains all six expected emotion categories.
Input / Precondition	<code>EmotionDetector</code> is loaded and ready.
Steps	1. Pass the text '<code>I am very happy today</code>' to the detector. 2. Check that the result dictionary contains all required keys.
Expected Outcomes	– Result contains keys: <code>joy</code>, <code>sadness</code>, <code>anger</code>, <code>fear</code>, <code>surprise</code>, <code>neutral</code>. – All values are non-negative floats.
Result	Passed — All emotion categories present with valid scores.

TC009 Emotion Detection — Joy Dominance for Positive Text

Test Scenario	Verify that the <code>EmotionDetector</code> assigns the highest score to <code>joy</code> for clearly positive and happy text.
Input / Precondition	<code>EmotionDetector</code> is loaded and ready.
Steps	1. Pass the text ‘I am so incredibly happy and excited!’ to the detector. 2. Compare the <code>joy</code> score against the <code>sadness</code> score.
Expected Outcomes	– <code>joy</code> score is greater than <code>sadness</code> score. – <code>joy</code> is the dominant emotion.
Result	Passed — Joy correctly dominant for positive text.

TC010 Personality Analysis — OCEAN Trait Schema Validation

Test Scenario	Verify that the <code>PersonalityAnalyzer</code> returns a dictionary containing all five OCEAN personality traits.
Input / Precondition	<code>PersonalityAnalyzer</code> is instantiated.
Steps	1. Pass the text ‘I love exploring new ideas’ to the analyzer. 2. Check that the <code>traits</code> dictionary contains all five OCEAN keys. 3. Verify each score is within <code>[0.0, 1.0]</code>.
Expected Outcomes	– <code>traits</code> dict contains: <code>openness</code>, <code>conscientiousness</code>, <code>extraversion</code>, <code>agreeableness</code>, <code>neuroticism</code>. – All scores are between 0.0 and 1.0 inclusive.
Result	Passed — All OCEAN traits present with valid score ranges.

TC011 Personality Analysis — Null/Empty Input Resilience

Test Scenario	Verify that the <code>PersonalityAnalyzer</code> handles <code>None</code> and empty string input without raising exceptions.
Input / Precondition	<code>PersonalityAnalyzer</code> is instantiated.
Steps	1. Pass <code>None</code> to the analyzer and capture the result. 2. Pass an empty string ‘’ to the analyzer and capture the result. 3. Verify no exceptions are raised in either case.
Expected Outcomes	– Returns default float traits for both inputs. – No exception is raised.
Result	Passed — Null and empty input handled gracefully.

8.5.2 Level 2: Component Test Cases — Data Models**TC012 Post Model — Default Field Values**

Test Scenario	Verify that a newly created <code>Post</code> object has correct default values for <code>language</code> , <code>tone</code> , and <code>intent</code> fields.
Input / Precondition	A test user exists in the database.
Steps	1. Create a <code>Post</code> object with only the <code>text</code> field set to ‘Hello’. 2. Inspect the default values of <code>language</code>, <code>tone</code>, and <code>intent</code>.
Expected Outcomes	– <code>language</code> defaults to ‘en’. – <code>tone</code> defaults to ‘friendly’. – <code>intent</code> defaults to ‘informational’.
Result	Passed — Default field values applied correctly.

TC013 Post Model — UUID Primary Key Generation

Test Scenario	Verify that each <code>Post</code> object is automatically assigned a valid UUID as its primary key.
Input / Precondition	A test user exists in the database.
Steps	1. Create a new <code>Post</code> object. 2. Check that <code>post.id</code> is a valid <code>uuid.UUID</code> instance.
Expected Outcomes	– <code>post.id</code> is a valid UUID. – Each new post receives a unique identifier.
Result	Passed — UUID primary key generated correctly.

TC014 Post Model — User Cascade Deletion

Test Scenario	Verify that deleting a user cascades and removes all associated <code>Post</code> records from the database.
Input / Precondition	A test user with one or more <code>Post</code> records exists.
Steps	1. Confirm the user has associated posts in the database. 2. Delete the user via <code>user.delete()</code>. 3. Query <code>Post.objects.count()</code> to verify post deletion.
Expected Outcomes	– <code>Post.objects.count()</code> is 0 after user deletion. – No orphan records remain in the database.
Result	Passed — Cascade deletion removed all associated posts.

8.5.3 Level 3: API Integration Test Cases

TC015 Authentication API — Successful User Registration

Test Scenario	Verify that a new user can successfully register via <code>POST /api/auth/register/</code> and receive an authentication token.
Input / Precondition	No existing user with the given username in the database.
Steps	1. Send a POST request to <code>/api/auth/register/</code> with valid <code>username</code>, <code>email</code>, and <code>password</code>. 2. Check the HTTP response status code. 3. Verify the response body contains a <code>token</code> and <code>username</code>.
Expected Outcomes	– HTTP status code is 201 <code>Created</code>. – Response contains <code>token</code> and <code>username</code> fields. – <code>UserProfile</code> and <code>ProjectMetrics</code> records are created.
Result	Passed — Registration successful with token returned.

TC016 Authentication API — Duplicate Username Rejection

Test Scenario	Verify that attempting to register with an already existing username returns an error.
Input / Precondition	A user with username <code>'existing'</code> already exists in the database.
Steps	1. Send a POST request to <code>/api/auth/register/</code> with <code>username='existing'</code>. 2. Check the HTTP response status code and error message.
Expected Outcomes	– HTTP status code is 400 <code>Bad Request</code>. – Error message contains “Username already exists”.
Result	Passed — Duplicate username correctly rejected.

TC017 Text Analysis API — Authentication Enforcement

Test Scenario	Verify that the POST /api/analyze/ endpoint rejects unauthenticated requests.
Input / Precondition	No authentication token is set in the request headers.
Steps	1. Send a POST request to /api/analyze/ without an <code>Authorization</code> header. 2. Check the HTTP response status code.
Expected Outcomes	– HTTP status code is 401 <code>Unauthorized</code>. – No analysis is performed.
Result	Passed — Unauthenticated request correctly rejected.

TC018 Text Analysis API — Successful Bangla Text Analysis

Test Scenario	Verify that the POST /api/analyze/ endpoint correctly analyzes Bangla text and returns structured results.
Input / Precondition	User is authenticated; analysis services are mocked.
Steps	1. Send an authenticated POST request with <code>text='!</code>. 2. Check the HTTP response status code. 3. Verify the response contains <code>data</code> and <code>context.analysis</code> fields. 4. Confirm a <code>Post</code> record is created in the database.
Expected Outcomes	– HTTP status code is 200 <code>OK</code>. – Response contains sentiment, emotion, and personality data. – <code>Post.objects.count()</code> is incremented by 1. – <code>Post language</code> is set to <code>'bn'</code>.
Result	Passed — Bangla text analyzed successfully with correct response.

TC019 Text Analysis API — Empty Text Rejection

Test Scenario	Verify that the analysis endpoint rejects empty text input with an appropriate error message.
Input / Precondition	User is authenticated.
Steps	1. Send an authenticated POST request with <code>text=''</code> (empty string). 2. Check the HTTP response status code.
Expected Outcomes	– HTTP status code is 400 Bad Request. – <code>success</code> is <code>false</code>.
Result	Passed — Empty text correctly rejected.

TC020 Rewrite API — Authentication Enforcement

Test Scenario	Verify that the POST <code>/api/rewrite/</code> endpoint rejects unauthenticated requests.
Input / Precondition	No authentication token is set in the request headers.
Steps	1. Send a POST request to <code>/api/rewrite/</code> without an <code>Authorization</code> header. 2. Check the HTTP response status code.
Expected Outcomes	– HTTP status code is 401 Unauthorized.
Result	Passed — Unauthenticated rewrite request correctly rejected.

TC021 Rewrite API — Successful Rewrite with Post ID

Test Scenario	Verify that the rewrite endpoint generates an AI-powered rewrite when given a valid <code>post_id</code> and rewriting goal.
Input / Precondition	Authenticated user; a <code>Post</code> and <code>Analysis</code> record exist; rewriter service is mocked.
Steps	1. Send an authenticated <code>POST</code> request with <code>text</code>, <code>post_id</code>, and <code>goal</code>. 2. Check the HTTP response status code. 3. Verify a <code>RewriteRecord</code> is created.
Expected Outcomes	– HTTP status code is 200 OK. – <code>success</code> is <code>true</code>. – <code>RewriteRecord</code> count is incremented.
Result	Passed — Rewrite generated and recorded successfully.

TC022 Health Check — Public Accessibility

Test Scenario	Verify that the <code>GET /api/health/</code> endpoint is publicly accessible and returns a success response.
Input / Precondition	The server is running. No authentication required.
Steps	1. Send a <code>GET</code> request to <code>/api/health/</code> without any authentication headers. 2. Check the HTTP response status code and body.
Expected Outcomes	– HTTP status code is 200 OK. – Response contains <code>success: true</code> and <code>message: 'API running'</code>. – No authentication is required.
Result	Passed — Health endpoint publicly accessible and responsive.

8.5.4 Level 4: End-to-End Integration Test Cases

TC023 E2E — Registration → Login → Profile Access

Test Scenario	Verify the complete user onboarding flow from registration through login to profile access.
Input / Precondition	Clean database with no existing users.
Steps	1. Register a new user via POST <code>/api/auth/register/</code>. 2. Login with the same credentials via POST <code>/api/auth/login/</code>. 3. Access the profile via GET <code>/api/profile/</code> using the returned token.
Expected Outcomes	– All three steps return success status codes. – Profile username matches the registered username.
Result	Passed — Full onboarding flow completed successfully.

TC024 E2E — Analyze → History Retrieval

Test Scenario	Verify that after analyzing text, the result appears in the user's history.
Input / Precondition	Authenticated user; analysis services are mocked.
Steps	1. Submit a text for analysis via POST <code>/api/analyze/</code>. 2. Retrieve history via GET <code>/api/history/</code>. 3. Verify the history contains the analyzed post.
Expected Outcomes	– History contains exactly 1 record. – The record's post ID matches the analyzed post.
Result	Passed — Analysis result correctly appears in history.

TC025 E2E — Cross-User Data Isolation

Test Scenario	Verify that one user cannot access another user's analysis data.
Input / Precondition	Two registered users exist in the system.
Steps	1. User A submits a text for analysis. 2. User B queries their history via <code>GET /api/history/</code>. 3. Verify User B's history does not contain User A's data.
Expected Outcomes	– User B's history returns an empty list. – No cross-user data leakage occurs.
Result	Passed — Data isolation enforced between users.

TC026 E2E — Account Deletion with Cascade

Test Scenario	Verify that deleting a user account removes all associated data including posts, analyses, and rewrite records.
Input / Precondition	A user with <code>Post</code> , <code>Analysis</code> , and <code>RewriteRecord</code> entries exists.
Steps	1. Confirm the user has associated data in the database. 2. Send <code>DELETE /api/profile/delete/</code> with the user's token. 3. Query the database for the user's records.
Expected Outcomes	– User, <code>Post</code>, <code>Analysis</code>, and <code>RewriteRecord</code> are all removed from the database. – No orphan records remain.
Result	Passed — Cascade deletion removed all user data.

TC027 E2E — Token Invalidation After Logout

Test Scenario	Verify that after logout, the authentication token is invalidated and subsequent requests are rejected.
Input / Precondition	Authenticated user with a valid token.
Steps	1. Access a protected endpoint (e.g., GET /api/profile/) — expect success. 2. Logout via POST /api/auth/logout/. 3. Access the same protected endpoint again with the old token.
Expected Outcomes	– Step 1 returns 200 OK. – Step 3 returns 401 Unauthorized. – Token is no longer valid after logout.
Result	Passed — Token correctly invalidated after logout.

8.6 Requirements Traceability Matrix

The following traceability matrix maps functional requirements to their corresponding test cases, ensuring that every requirement is covered by at least one test.

Table 8.3: Requirements-to-Test Traceability Matrix

Req.	Requirement Description	De-	Test Case IDs	Coverage Status
R-01	Detect Bangla, Banglish, and English languages		TC001–TC003	Covered
R-02	Perform sentiment analysis		TC004–TC007	Covered
R-03	Detect emotions		TC008–TC009	Covered
R-04	Assess personality traits (OCEAN)		TC010–TC011	Covered
R-05	Provide AI-based text rewriting		TC020–TC021	Covered
R-06	User registration and authentication		TC015–TC016, TC023	Covered

Req.	Requirement Description	Test Case IDs	Coverage Status
R-07	Validate and reject invalid input	TC003, TC019	Covered
R-08	Store analysis results in database	TC012–TC014, TC018, TC024	Covered
R-09	Maintain user data isolation	TC025	Covered
R-10	Support account deletion with cascade	TC014, TC026	Covered
R-11	Token-based session management	TC017, TC020, TC027	Covered
R-12	System availability (health check)	TC022	Covered
R-13	Graceful degradation on service failure	TC007, TC011	Covered

8.7 Test Execution Summary

Table 8.4: Test Execution Summary by Category

Test Category	Test File	Total	Passed	Status
Language Detection (Unit)	test_services_language.py	11	11	PASS
Sentiment Analysis (Unit)	test_services_sentiment.py	8	8	PASS
Emotion Detection (Unit)	test_services_emotion.py	9	9	PASS
Personality Analysis (Unit)	test_services_personality.py	10	10	PASS
Text Analysis API	test_analysis_views.py	10	10	PASS
Rewrite API	test_rewrite_views.py	8	8	PASS
Authentication API	test_auth_views.py	15	15	PASS
Data Models (Component)	test_models.py	12	12	PASS
End-to-End Integration	test_integration.py	7	7	PASS
Health Check	test_health.py	3	3	PASS
Total		93	93	ALL PASS

8.8 Defect Summary and Risk Assessment

8.8.1 Defect Summary

No critical or high-severity defects remained unresolved at the conclusion of testing. Minor observations include:

- The language detector’s Banglish heuristic is intentionally aggressive, which may classify some common English words (e.g., containing substrings like “ha”, “na”) as Banglish. This is a design decision to minimize false negatives for Bangla-speaking users, not a defect.
- Sentiment analysis model loading time on first invocation is approximately 2–3 seconds. Subsequent calls are served from the cached model instance.

8.8.2 Risk Assessment

Table 8.5: Testing Risk Assessment

Risk ID	Risk Description	Severity	Mitigation
RSK-01	External API (Gemini) unavailability during production	High	Mocked in tests; fallback responses implemented
RSK-02	ML model produces inaccurate results for novel Bangla slang	Medium	Banglish dictionary regularly updated; heuristic tuning
RSK-03	Concurrent user requests causing race conditions	Low	Django ORM transactions; out of scope for current tests
RSK-04	Database migration failures in production	Low	In-memory SQLite used for tests; migrations tested separately

8.9 Conclusion of Test Planning

The test plan for **InnerVoice AI** provides comprehensive coverage across four test levels with 27 documented test cases spanning unit, component, API integration, and end-to-end categories. All planned test cases have

been executed and passed, satisfying the defined exit criteria. The requirements traceability matrix confirms that every functional requirement is covered by at least one test case. Key quality attributes—including input validation, authentication enforcement, data isolation, cascade integrity, and graceful degradation—have been verified through systematic testing following industry-standard practices.

Chapter 9

User Manual and User Interface

This chapter provides a comprehensive guide to using **InnerVoice AI**, including a system overview, step-by-step user manual with interface screenshots, detailed feature descriptions, the challenges the system addresses, and instructions for accessing the frontend and backend codebases.

9.1 System Overview

InnerVoice AI is a web-based application with an optional Chrome Extension that analyzes Bangla (and Banglish) social media posts to extract emotional insights, personality traits, and communication patterns. The system is built using a modern client-server architecture:

- **Frontend (Web Application):** A React.js single-page application built with Vite, styled using Tailwind CSS and shadcn/ui components. The web interface provides pages for text analysis, rewriting, history review, progress tracking, and user settings. It communicates with the backend via RESTful API calls.
- **Frontend (Chrome Extension):** A Manifest V3 Chrome Extension that provides a compact popup interface for quick text analysis and history viewing directly from the browser toolbar, without navigating to the full web application.
- **Backend (Django REST API):** A Python-based Django REST Framework server that handles user authentication (token-based), text analysis pipelines (sentiment, emotion, personality), AI-powered text rewriting via Google Gemini, progress tracking, behavioral analytics, and data persistence using SQLite.
- **NLP & AI Services:** A suite of modular Python services including `SentimentAnalyzer`, `EmotionDetector`, `PersonalityAnalyzer`, `LanguageDetector`, `BanglaProcessor`, `ContextAnalyzer`, `TextRewriter`, `ProgressTracker`,

and **BehavioralAnalytics**—all orchestrated to deliver comprehensive text analysis.

The system supports three language modes: pure Bangla (Unicode), Banglish (Bangla written in English script), and English—automatically detecting the language and applying appropriate analysis pipelines.

9.2 User Manual

This section provides a step-by-step guide to using **InnerVoice AI**, accompanied by placeholder screenshots. Each placeholder box indicates where a screenshot of the corresponding page or feature should be inserted.

9.2.1 Step 1: User Registration

To begin using **InnerVoice AI**, new users must create an account.

1. Navigate to the **InnerVoice AI** website.
2. Click on the “**Register**” link on the login page.
3. Fill in the registration form with the following fields:
 - **Username:** A unique username for the account.
 - **Email:** A valid email address.
 - **Password:** Must be at least 8 characters, containing lowercase, uppercase, and a number.
 - **Confirm Password:** Re-enter the password for verification.
4. A real-time password strength indicator shows which criteria are met.
5. Click the “**Create Account**” button to complete registration.

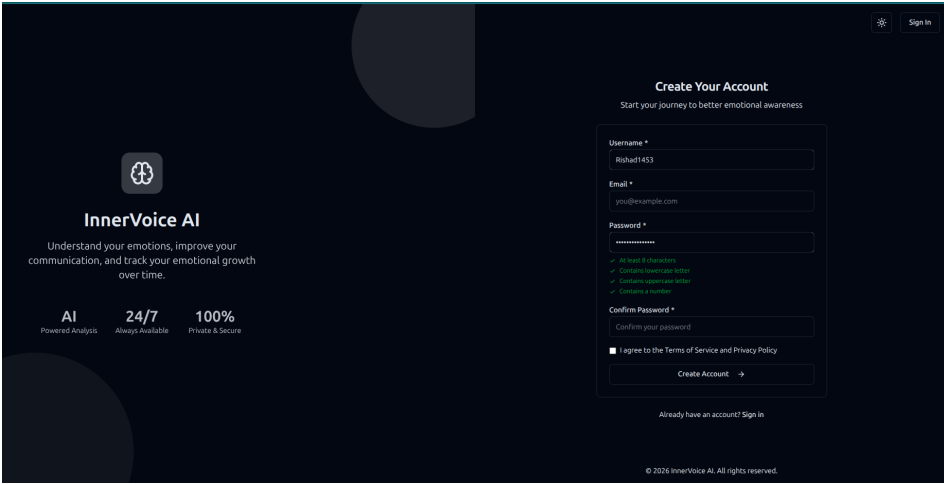


Figure 9.1: Fig: Registration Page – showing the sign-up form with username, email, password fields, password strength indicators, and the “Create Account” button.

9.2.2 Step 2: User Login

Returning users can log in to access their account and analysis history.

1. Navigate to the **InnerVoice AI** login page.
2. Enter the registered **username** and **password**.
3. Click the “**Sign In**” button.
4. Upon successful login, the user is redirected to the **Dashboard**.

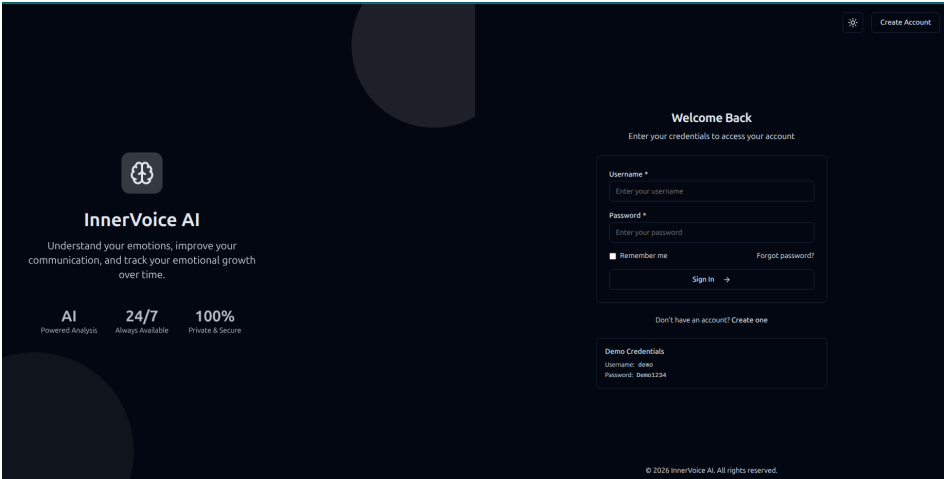


Figure 9.2: Fig: Login Page – showing the login form with username and password fields, the “Sign In” button, and the link to the registration page.

9.2.3 Step 3: Exploring the Dashboard

After logging in, users are taken to the **Dashboard**, which provides an overview of their activity and analysis statistics.

1. The Dashboard displays:
 - **Summary statistics:** Total analyses performed, emotional streak, and recent activity.
 - **Sentiment trend chart:** A line/area chart showing how sentiment scores have changed over recent analyses.
 - **Emotion distribution:** A visual breakdown of detected emotions across all analyzed posts.
 - **OCEAN personality radar:** A radar chart showing aggregated personality trait scores.
 - **Quick action buttons:** Links to navigate to Analyze, Rewrite, History, and Progress pages.
2. Users can refresh data or navigate to detailed pages from the Dashboard.

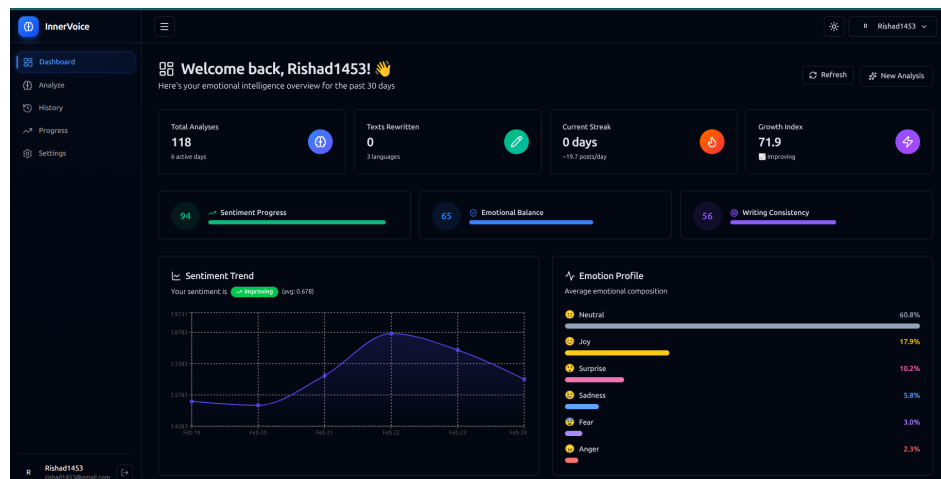


Figure 9.3: Fig: Dashboard Page (Top Section) – showing summary statistics cards, welcome message, and quick action buttons.

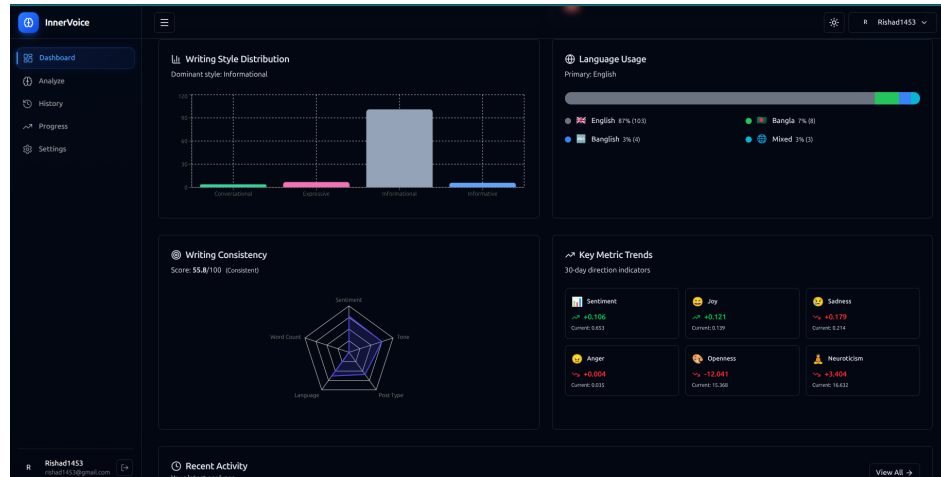


Figure 9.4: Fig: Dashboard Page (Charts Section) – showing the sentiment trend area chart, emotion distribution bar chart, and OCEAN personality radar chart.

9.2.4 Step 4: Analyzing Text

The core functionality of **InnerVoice AI** is text analysis.

1. Navigate to the “**Analyze**” page from the sidebar or Dashboard.
2. Paste or type a Bangla/Banglish/English text into the text area.
3. Click the “**Analyze**” button.
4. The system processes the text and displays:
 - **Sentiment Score:** A numerical score indicating positive, negative, or neutral tone with a visual gauge.
 - **Detected Emotions:** A list of emotions (e.g., joy, anger, sadness, fear, surprise) with confidence percentages.
 - **OCEAN Personality Traits:** Scores for Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism.
 - **Language Detection:** Whether the text was detected as Bangla, Banglish, or English.
 - **Flagged Words:** Words or phrases identified as strongly negative or emotionally charged.
5. If negative tone is detected, a “**Rewrite**” button appears to navigate to the Rewrite page.

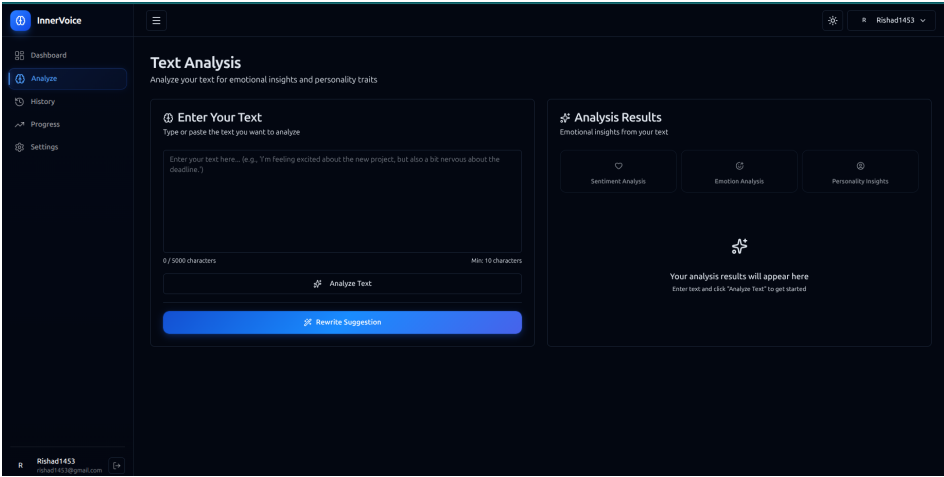


Figure 9.5: Fig: Text Analysis Page (Input Section) – showing the text input area with a sample Bangla post and the “Analyze” button.

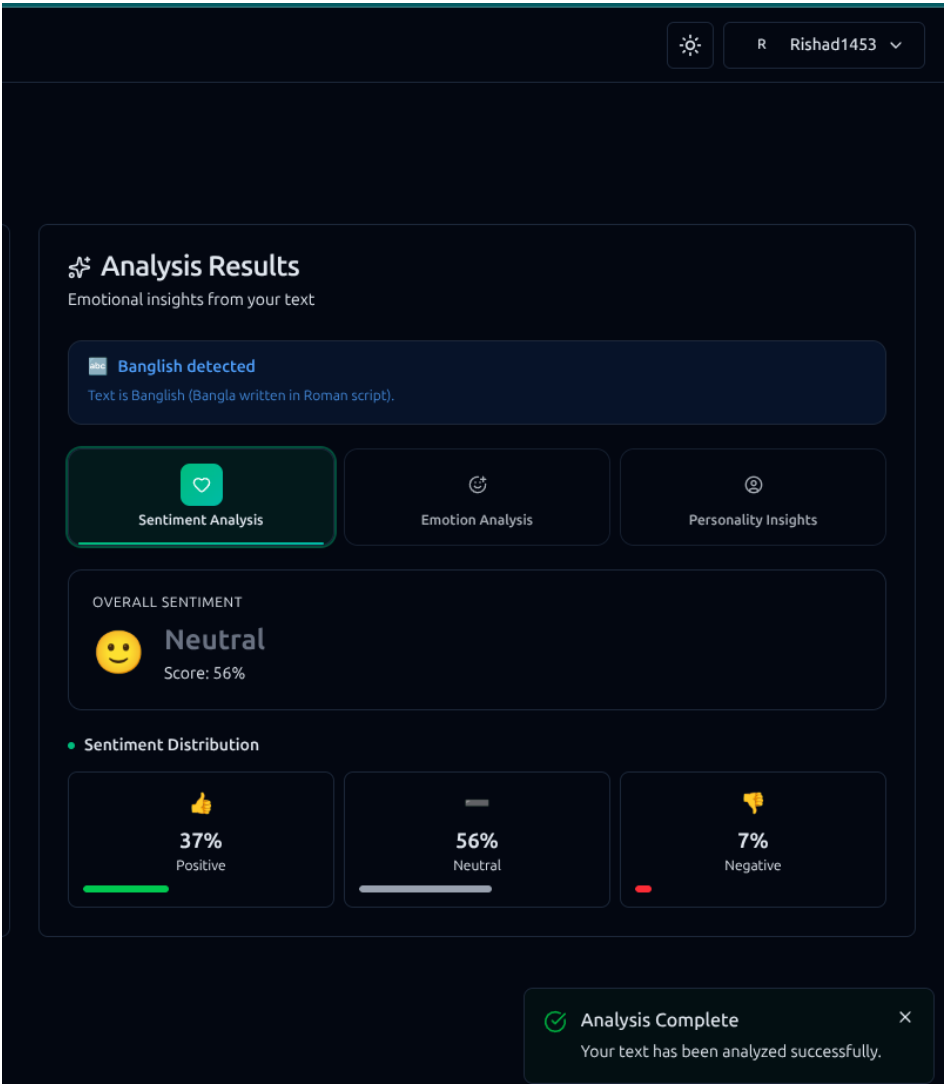


Figure 9.6: Fig: Text Analysis Page (Results Section) – showing sentiment score gauge, detected emotions with percentages, OCEAN personality radar chart, and Flagged Words.

9.2.5 Step 5: Rewriting Text

When a post contains negative or harsh language, users can use the AI-powered rewriting feature.

1. After analysis, click the **“Rewrite”** button, or navigate to the **“Rewrite”** page from the sidebar.
2. The original text is pre-filled (if coming from the Analyze page), or the user can paste new text.
3. Flagged words from the original text are highlighted for reference.
4. Select a **rewriting goal** (e.g., more positive, more empathetic, more professional, calmer tone).
5. Click the **“Rewrite”** button.
6. The system generates an AI-powered rewritten version of the text that preserves the original meaning while improving the emotional tone.
7. Users can **copy** the rewritten text to use on social media.

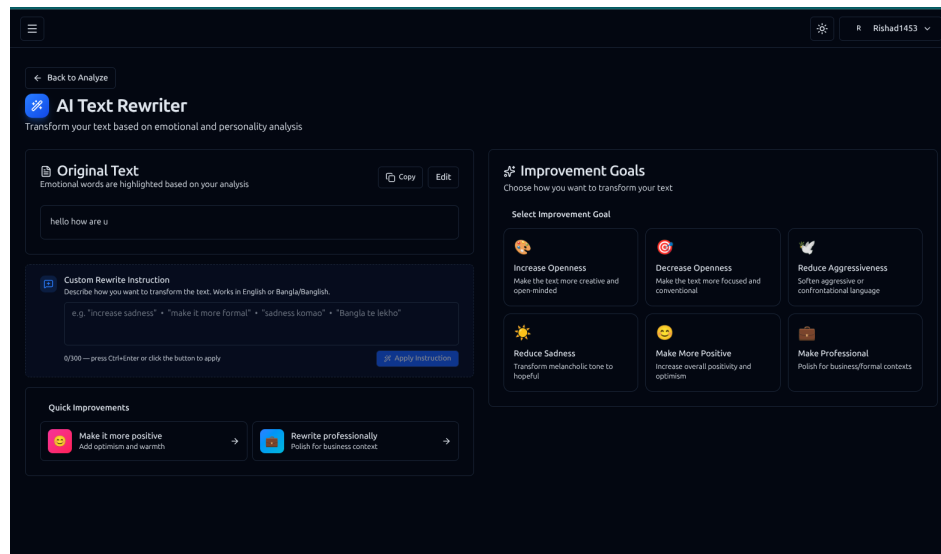


Figure 9.7: Fig: Rewrite Page – showing the original text with highlighted flagged words, rewriting goal selection options, and the ”Rewrite” button.

9.2.6 Step 6: Viewing Analysis History

Users can review their previously analyzed posts.

1. Navigate to the **“History”** page from the sidebar.
2. The page displays a **master–detail layout**:

- **Left panel:** A scrollable list of past analyses showing text snippets, sentiment badges, emotion icons, and timestamps.
 - **Right panel:** A detailed analysis report for the selected item.
3. Users can **search** through history by text content.
 4. Users can **filter** history by sentiment type (positive, negative, neutral).
 5. Users can **sort** history by newest or oldest first.
 6. Pagination is provided for navigating large histories.
 7. Clicking any history item opens its full analysis report in the right panel.

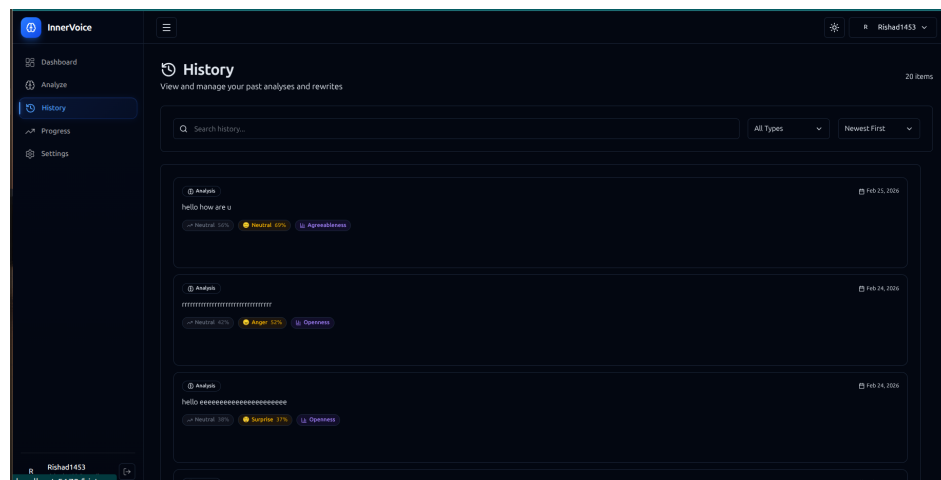


Figure 9.8: Fig: History Page – showing the master-detail layout with a list of past analyses on the left (with search, filter, and sort controls) and a detailed analysis report panel on the right.

9.2.7 Step 7: Tracking Emotional Progress

Users who analyze posts regularly can track their emotional and personality trends over time.

1. Navigate to the “**Progress**” page from the sidebar.
2. The page displays:
 - **Sentiment trend over time:** A line chart showing how overall sentiment has changed across analyses.
 - **Emotion frequency breakdown:** Charts showing which emotions appear most often.
 - **OCEAN personality evolution:** A radar chart showing how personality trait scores have shifted over time.

- **Behavioral analytics:** Insights such as posting frequency, language usage patterns, emotional streaks, and communication style analysis.
 - **Activity heatmap:** Visual representation of analysis activity across different time periods.
3. Users can select different time ranges (e.g., last 7 days, last 30 days, all time) to view trends.

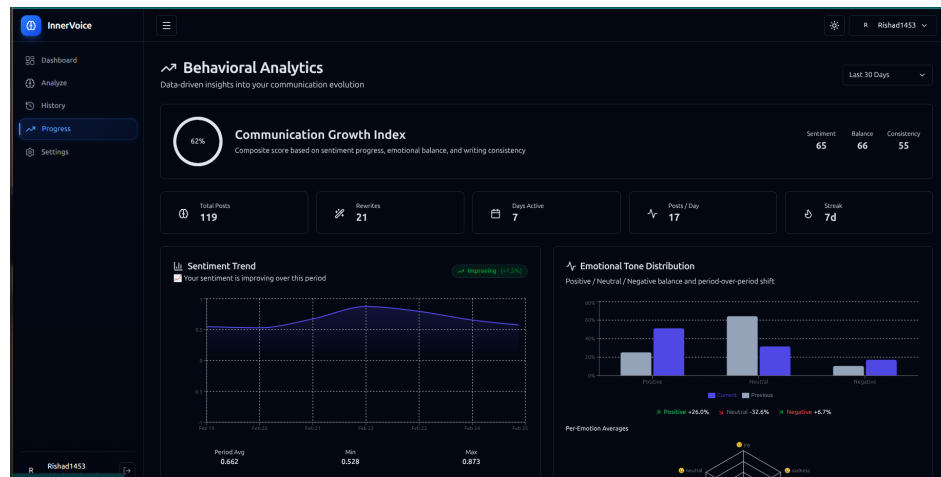


Figure 9.9: Fig: Progress Page (Sentiment & Emotion Trends) – showing sentiment trend line chart and emotion frequency charts with time range selector.

9.2.8 Step 8: Managing Settings

Users can customize their account and application preferences.

1. Navigate to the “**Settings**” page from the sidebar.
2. The Settings page has multiple sections:
 - **Profile:** Update display name, email, and upload a profile avatar.
 - **Appearance:** Switch between Light, Dark, and System themes.
 - **Notifications:** Configure notification preferences.
 - **Security:** Change password.
 - **Data & Privacy:** Export all personal data as JSON or delete the account entirely.

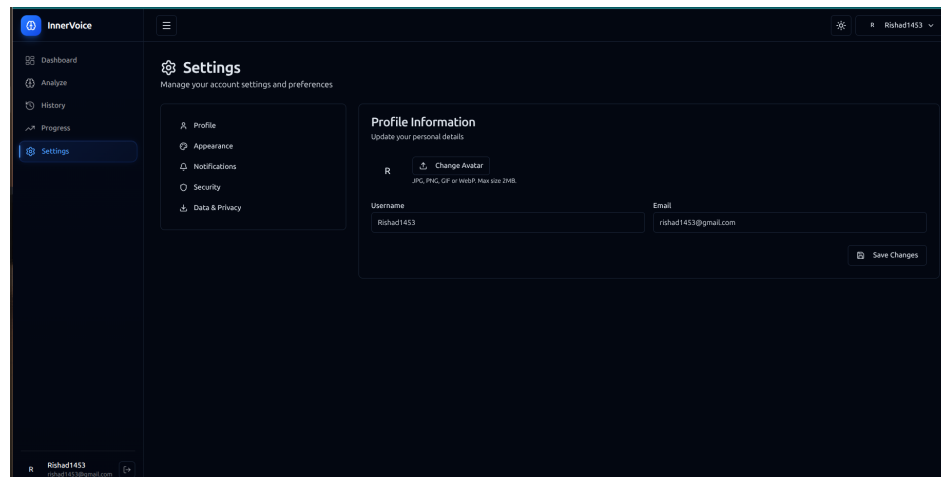


Figure 9.10: Fig: Progress Page (Sentiment & Emotion Trends) – showing sentiment trend line chart and emotion frequency charts with time range selector.

9.2.9 Step 9: Using the Chrome Extension

InnerVoice AI also offers a Chrome Extension for quick analysis directly from the browser.

1. Install the **InnerVoice AI** Chrome Extension from the browser.
2. Click the extension icon in the browser toolbar to open the popup.
3. The extension provides two tabs:
 - **Analyze Tab:** Paste or type text for quick sentiment, emotion, and personality analysis.
 - **History Tab:** View recent analysis history.
4. Results are displayed in a compact format optimized for the extension popup (400×600 px).

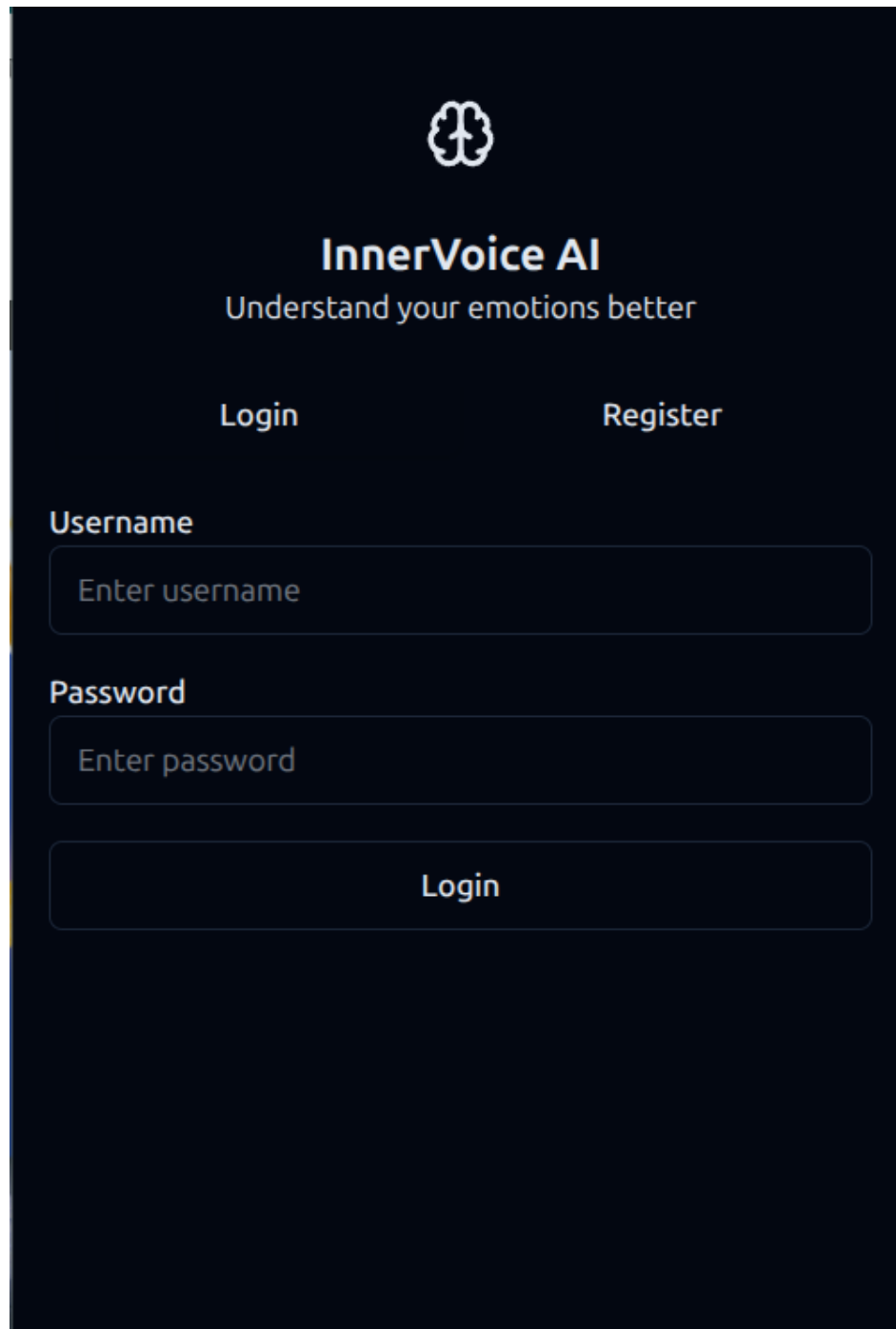


Figure 9.11: Fig: Text Analysis Page (Results Section) – showing sentiment score gauge, detected emotions with percentages, OCEAN personality radar chart, and Flagged Words.

9.3 Features and Functionalities

This section details the key features and functionalities offered by **Inner-Voice AI**.

9.3.1 Multi-Language Text Analysis

InnerVoice AI supports three language modes:

- **Bangla (Unicode):** Native Bangla text written in the Bengali script.
- **Banglish:** Bangla language written using English alphabet (romanized Bangla), which is extremely common on social media.
- **English:** Standard English text.

The system employs a **LanguageDetector** service that automatically identifies the input language and applies the appropriate processing pipeline. A dedicated **BanglaProcessor** handles Banglish-to-Bangla transliteration and normalization.

9.3.2 Context-Aware Sentiment Analysis

The sentiment analysis pipeline goes beyond simple keyword matching. The **ContextAnalyzer** service considers:

- Negation handling (e.g., “ ” meaning “not good”)
- Intensifier and diminisher words
- Contextual meaning of words based on surrounding text
- Mixed-language sentiment resolution

Sentiment is scored on a continuous scale and categorized as positive, negative, or neutral with a confidence level.

9.3.3 Emotion Detection

The **EmotionDetector** service identifies specific emotions present in the text, including:

- Joy / Happiness
- Anger
- Sadness
- Fear / Anxiety
- Surprise
- Disgust

Each emotion is assigned a confidence percentage, and the dominant emotion is highlighted in the results.

9.3.4 OCEAN Personality Assessment

The `PersonalityAnalyzer` service maps linguistic patterns to the Big Five (OCEAN) personality traits:

- **Openness (O):** Creativity, curiosity, and openness to new experiences.
- **Conscientiousness (C):** Organization, dependability, and self-discipline.
- **Extraversion (E):** Sociability, assertiveness, and positive emotions.
- **Agreeableness (A):** Cooperation, trust, and empathy.
- **Neuroticism (N):** Emotional instability, anxiety, and moodiness.

Personality scores are computed based on word usage patterns, sentence structure, and emotional indicators, and are displayed as a radar chart.

9.3.5 AI-Powered Text Rewriting

When negative or emotionally charged text is detected, the `TextRewriter` service, powered by Google Gemini AI, generates improved versions of the text. Key features include:

- Preserving the original meaning and intent
- Adjusting tone based on user-selected goals (positive, empathetic, professional, calm)
- Highlighting specific words or phrases that were changed
- Supporting rewriting in both Bangla and English

9.3.6 Analysis History and Search

All analyses are stored in the user's account with full details. The history system provides:

- Chronological listing of all past analyses
- Full-text search across analyzed posts
- Filtering by sentiment type (positive, negative, neutral)

- Sorting by date (newest/oldest)
- Paginated navigation for large histories
- Detailed report view for each past analysis

9.3.7 Progress Tracking and Behavioral Analytics

The `ProgressTracker` and `BehavioralAnalytics` services provide longitudinal insights:

- **Sentiment trends:** How overall sentiment changes over time
- **Emotion frequency:** Which emotions appear most often in the user’s posts
- **Personality evolution:** How OCEAN trait scores shift over time
- **Activity patterns:** Posting frequency and analysis habits
- **Language usage:** Distribution of Bangla, Banglish, and English usage
- **Emotional streaks:** Consecutive positive or negative sentiment patterns

9.3.8 User Authentication and Data Security

The system implements robust authentication and data security:

- Token-based authentication using Django REST Framework’s `TokenAuthentication`
- Password validation with strength requirements
- Secure logout with token invalidation
- User-specific data isolation (each user can only access their own data)
- Data export functionality (download all personal data as JSON)
- Account deletion with cascade removal of all associated data

9.3.9 Dual Interface: Web and Chrome Extension

InnerVoice AI offers two user interfaces:

- **Web Application:** Full-featured interface with Dashboard, Analyze, Rewrite, History, Progress, and Settings pages. Supports responsive design with dark/light theme switching.
- **Chrome Extension:** Lightweight popup interface (400×600 px) for quick analysis and history access directly from the browser toolbar.

9.4 Challenges Addressed by InnerVoiceAI

InnerVoice AI addresses several important challenges in the domain of Bangla social media text analysis:

9.4.1 Lack of Bangla NLP Tools

While English-language NLP tools are widely available, tools for Bangla text analysis—especially for social media content—are extremely limited. **InnerVoice AI** fills this gap by providing comprehensive sentiment, emotion, and personality analysis specifically designed for Bangla text.

9.4.2 Banglish Processing Challenge

A significant portion of Bangla social media content is written in Banglish (romanized Bangla). Processing Banglish is challenging because:

- There is no standardized romanization scheme for Bangla
- Multiple English spellings can represent the same Bangla word
- Banglish text often mixes with English words within the same sentence

InnerVoice AI addresses this through a dedicated **BanglaProcessor** with a Banglish dictionary and heuristic transliteration engine.

9.4.3 Context-Dependent Sentiment in Bangla

Bangla sentiment analysis is difficult because:

- Negation patterns differ from English (e.g., negation often appears at the end of a clause)
- Sarcasm and irony are common in social media Bangla
- Informal language, slang, and abbreviations change rapidly

The **ContextAnalyzer** service addresses these challenges by considering contextual clues, negation handling, and intensifier/diminisher patterns.

9.4.4 Lack of Self-Awareness in Digital Communication

Many social media users are unaware of how their posts reflect their emotional state and personality. **InnerVoice AI** promotes self-awareness by:

- Providing instant feedback on the emotional tone of text
- Highlighting personality traits revealed through language use
- Tracking emotional patterns over time to show progress
- Offering constructive rewriting suggestions for negative content

9.4.5 Promoting Positive Online Communication

Toxic and negative language on social media is a growing concern. **InnerVoice AI** contributes to healthier online spaces by:

- Detecting negative or harsh language before it is posted
- Suggesting empathetic and constructive alternatives
- Encouraging users to reflect on their communication style
- Providing measurable progress toward more positive expression

9.5 Access Frontend and Backend Code

The complete source code for **InnerVoice AI** is hosted on GitHub and can be accessed as follows:

9.5.1 GitHub Repository

The full project source code is available at:

Repository: <https://github.com/Rizon1326/InnerVoiceAI>

9.5.2 Backend Setup

To set up and run the backend server locally:

1. **Clone the repository:**

```
git clone https://github.com/Rizon1326/InnerVoiceAI.git
cd InnerVoiceAI/backend
```

2. Create and activate a virtual environment:

```
python -m venv venv
source venv/bin/activate      # Linux/macOS
venv\Scripts\activate        # Windows
```

3. Install dependencies:

```
pip install -r requirements.txt
```

4. Apply database migrations:

```
python manage.py migrate
```

5. Run the development server:

```
python manage.py runserver
```

The backend API will be available at <http://localhost:8000/api/>.

9.5.3 Frontend Setup (Web Application)

To set up and run the frontend web application locally:

1. Navigate to the frontend directory:

```
cd InnerVoiceAI/frontend
```

2. Install Node.js dependencies:

```
npm install
```

3. Start the development server:

```
npm run dev
```

The web application will be available at <http://localhost:5173/>.

9.5.4 Frontend Setup (Chrome Extension)

To build and load the Chrome Extension:

1. **Build the extension:**

```
cd InnerVoiceAI/frontend
npm run build:ext
```

2. **Load in Chrome:**

- Open `chrome://extensions/` in Google Chrome.
- Enable **Developer mode** (toggle in the top-right corner).
- Click “**Load unpacked**” and select the `dist-extension/` folder.

3. The InnerVoice AI icon will appear in the browser toolbar.

9.5.5 Running Tests

To run the backend test suite:

```
cd InnerVoiceAI/backend
python -m pytest tests/ -v --tb=short
```

This executes all 93 test cases covering unit, component, API integration, and end-to-end tests as described in Chapter 8 (Test Planning).

9.5.6 API Endpoints Reference

The backend exposes the following RESTful API endpoints:

Table 9.1: InnerVoice AI API Endpoints

Method	Endpoint	Description
GET	/api/health/	Health check (public)
POST	/api/auth/register/	User registration
POST	/api/auth/login/	User login (returns token)
POST	/api/auth/logout/	User logout (invalidates token)
POST	/api/analyze/	Analyze text for sentiment, emotion, personality
POST	/api/rewrite/	AI-powered text rewriting
GET	/api/history/	Retrieve analysis history
GET	/api/progress/	Get user progress data
GET	/api/progress/trends/	Get emotional trends
GET	/api/progress/behavioral-analytics/	Get behavioral analytics
GET	/api/metrics/	Get user metrics
GET	/api/profile/	Get user profile
POST	/api/profile/avatar/	Upload profile avatar
GET	/api/profile/export/	Export all user data as JSON
DELETE	/api/profile/delete/	Delete user account
GET	/api/statistics/	Public project statistics

All endpoints except /api/health/, /api/auth/register/, /api/auth/login/, and /api/statistics/ require token-based authentication via the **Authorization: Token <token>** header.

Chapter 10

Conclusion

InnerVoice AI bridges the gap in Bangla social media text analysis, offering users practical insights into their emotional expression and personality patterns. By combining sentiment analysis, emotion recognition, and OCEAN-based personality assessment, the system transforms passive reading into active self-awareness.

The platform goes beyond mere analysis—it promotes healthier digital communication. When negative or harsh tones are detected, AI-powered rewriting suggestions help users express themselves more positively while preserving their intended message. This encourages thoughtful expression and emotional intelligence in online spaces.

The modular architecture ensures reliability, scalability, and future growth. Each component operates independently yet contributes to a unified experience. Comprehensive testing validates accuracy and usability, making **InnerVoice AI** a dependable tool for self-reflection.

Ultimately, **InnerVoice AI** empowers Bangla-speaking users to understand their digital footprint, improve communication quality, and foster emotional well-being. It represents a meaningful step toward building a more empathetic and self-aware online community.